

BFS Traversal

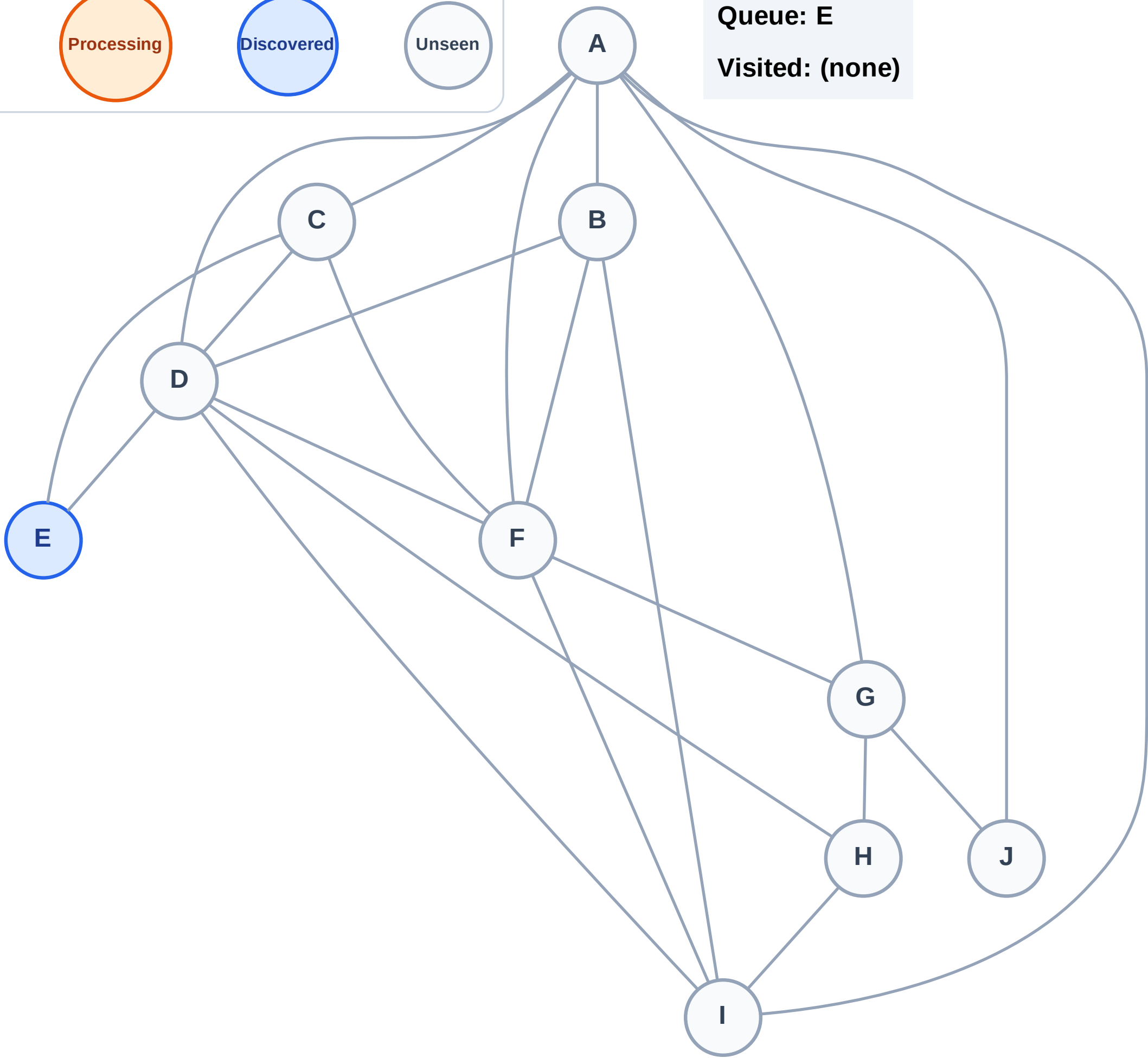
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

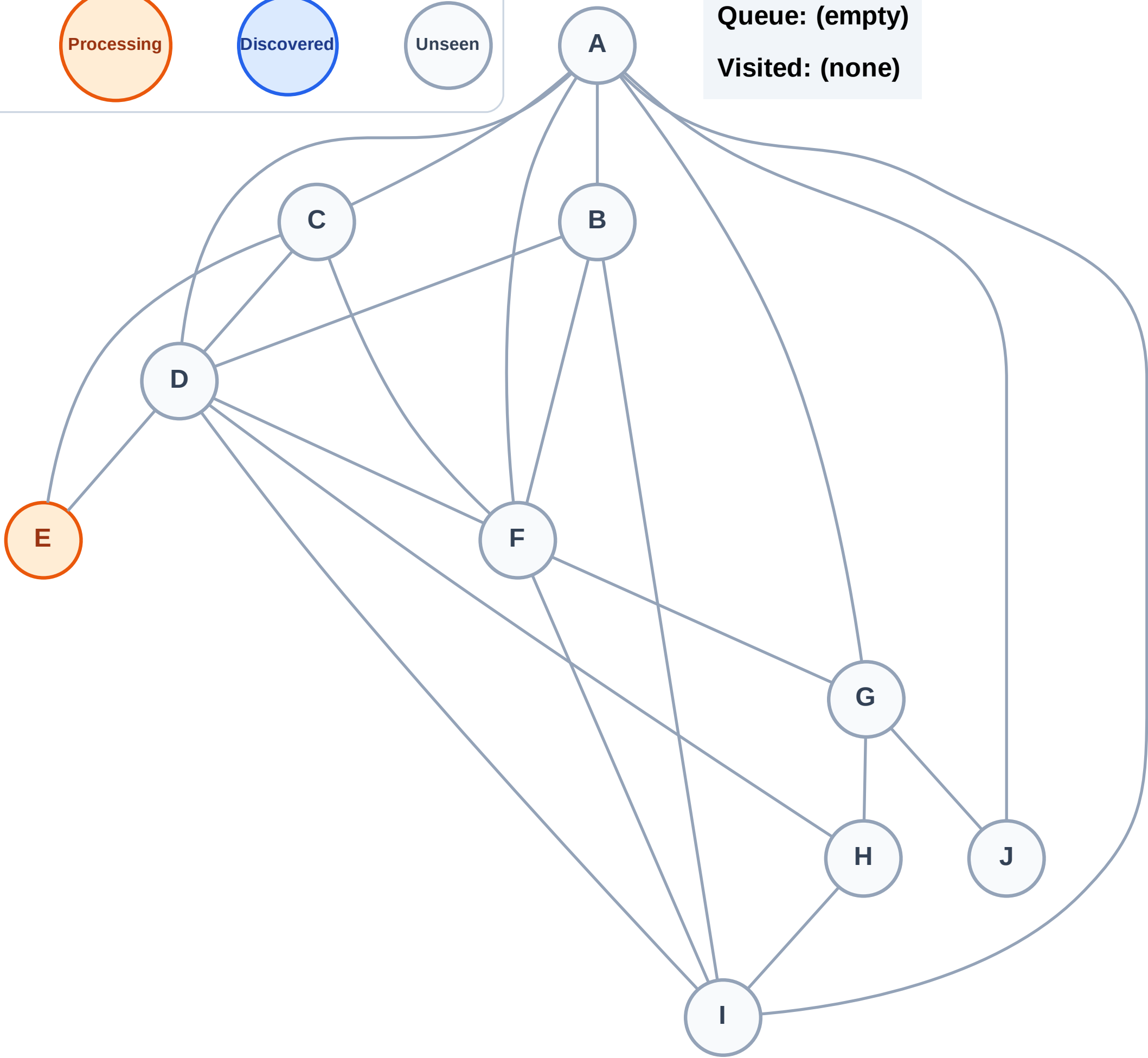
Step 2: Process node E

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

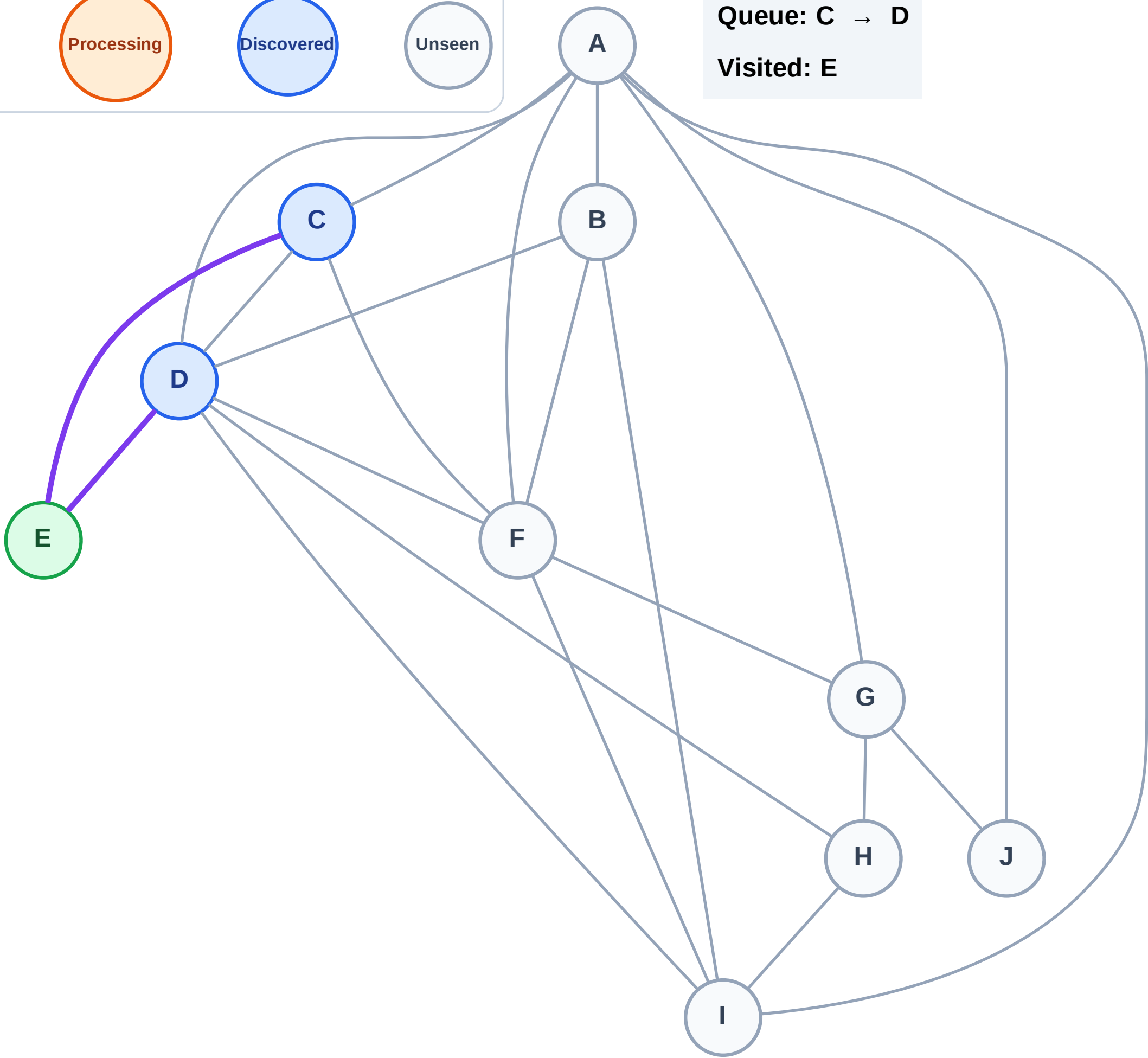
Step 3: Discover C, D from E

Legend



Queue: C → D

Visited: E



BFS Traversal

Step 4: Process node C

Legend

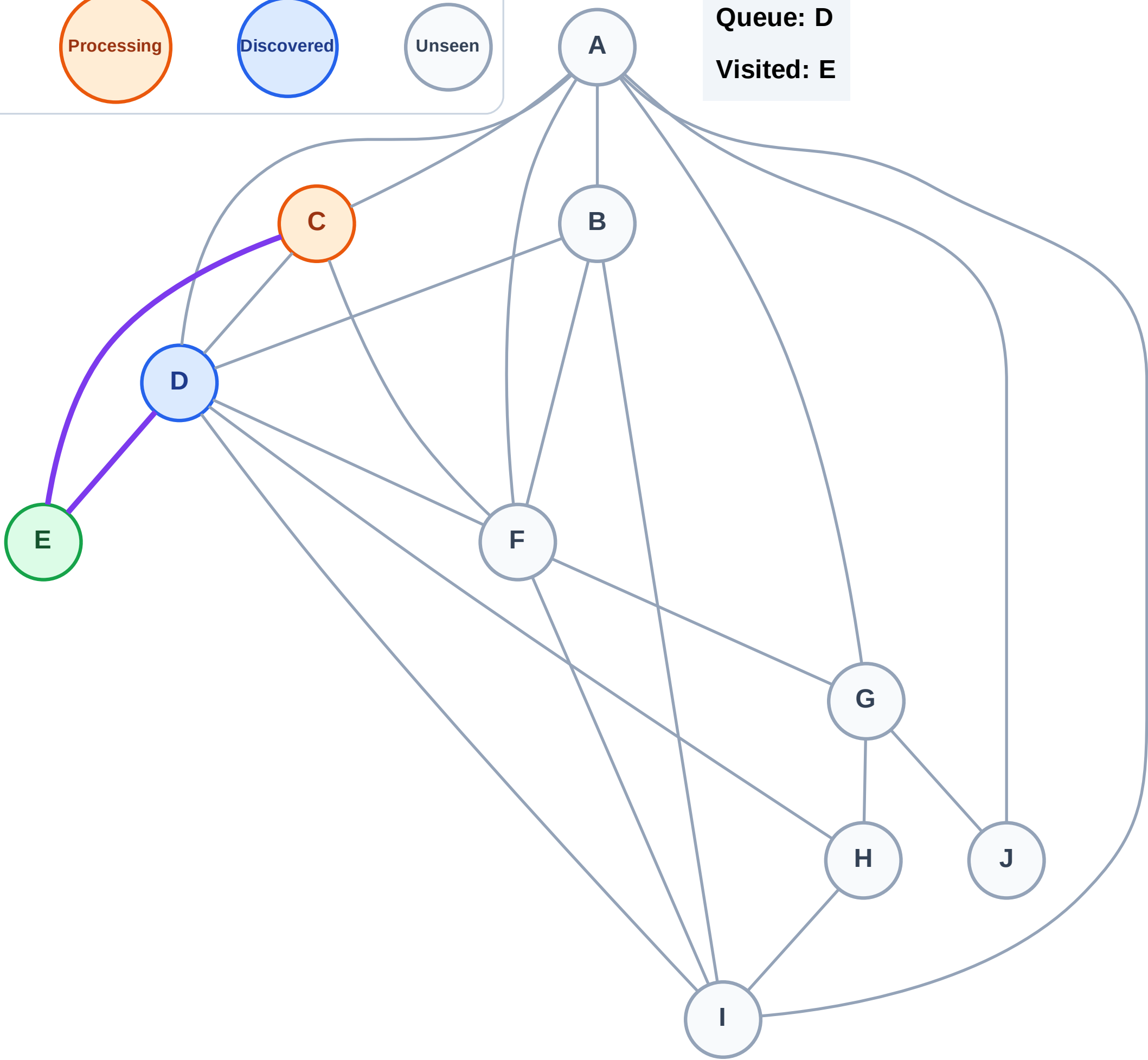
Complete

Processing

Discovered

Unseen

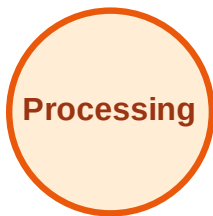
Queue: D
Visited: E



BFS Traversal

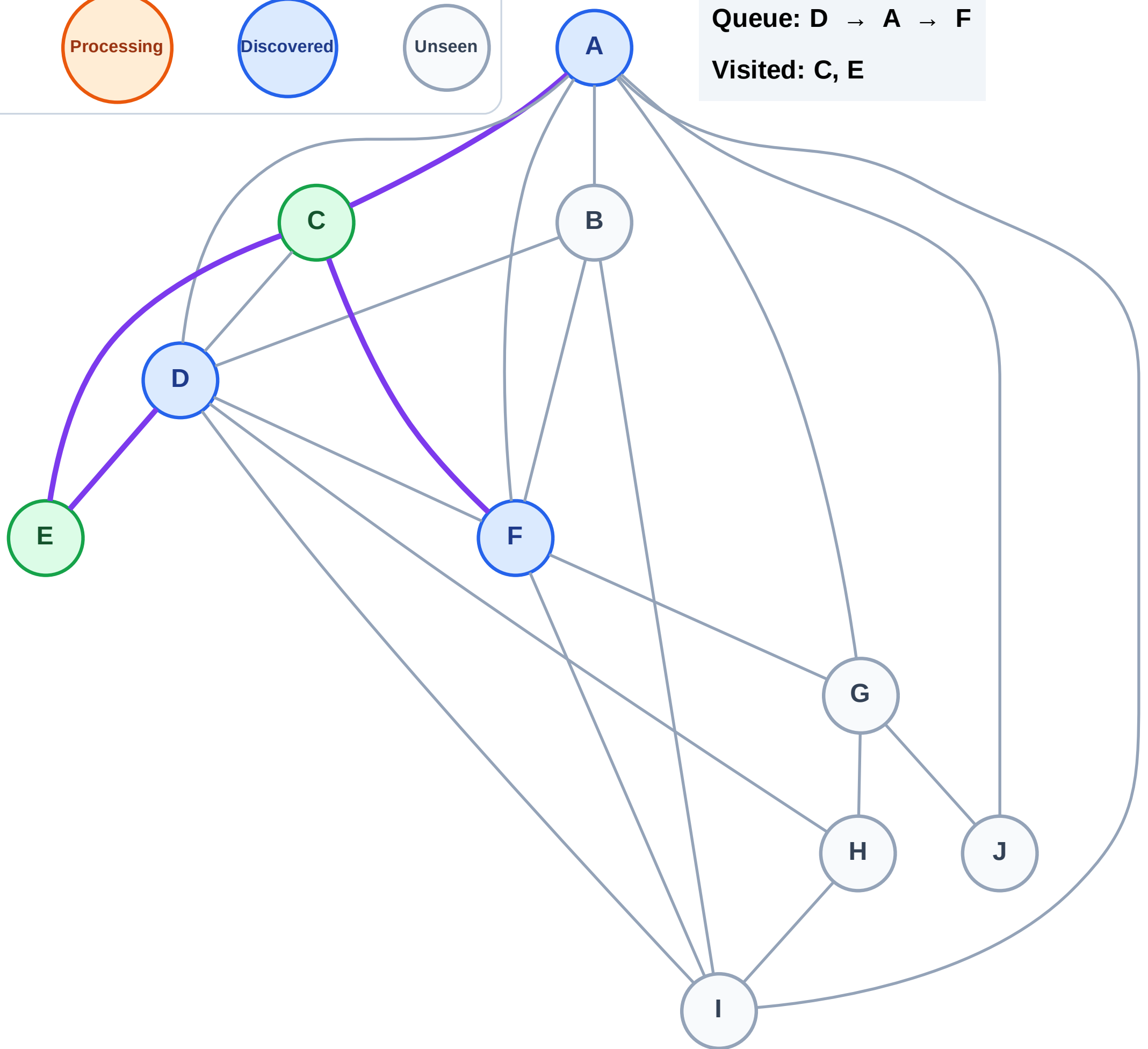
Step 5: Discover A, F from C

Legend



Queue: D → A → F

Visited: C, E



BFS Traversal

Step 6: Process node D

Legend

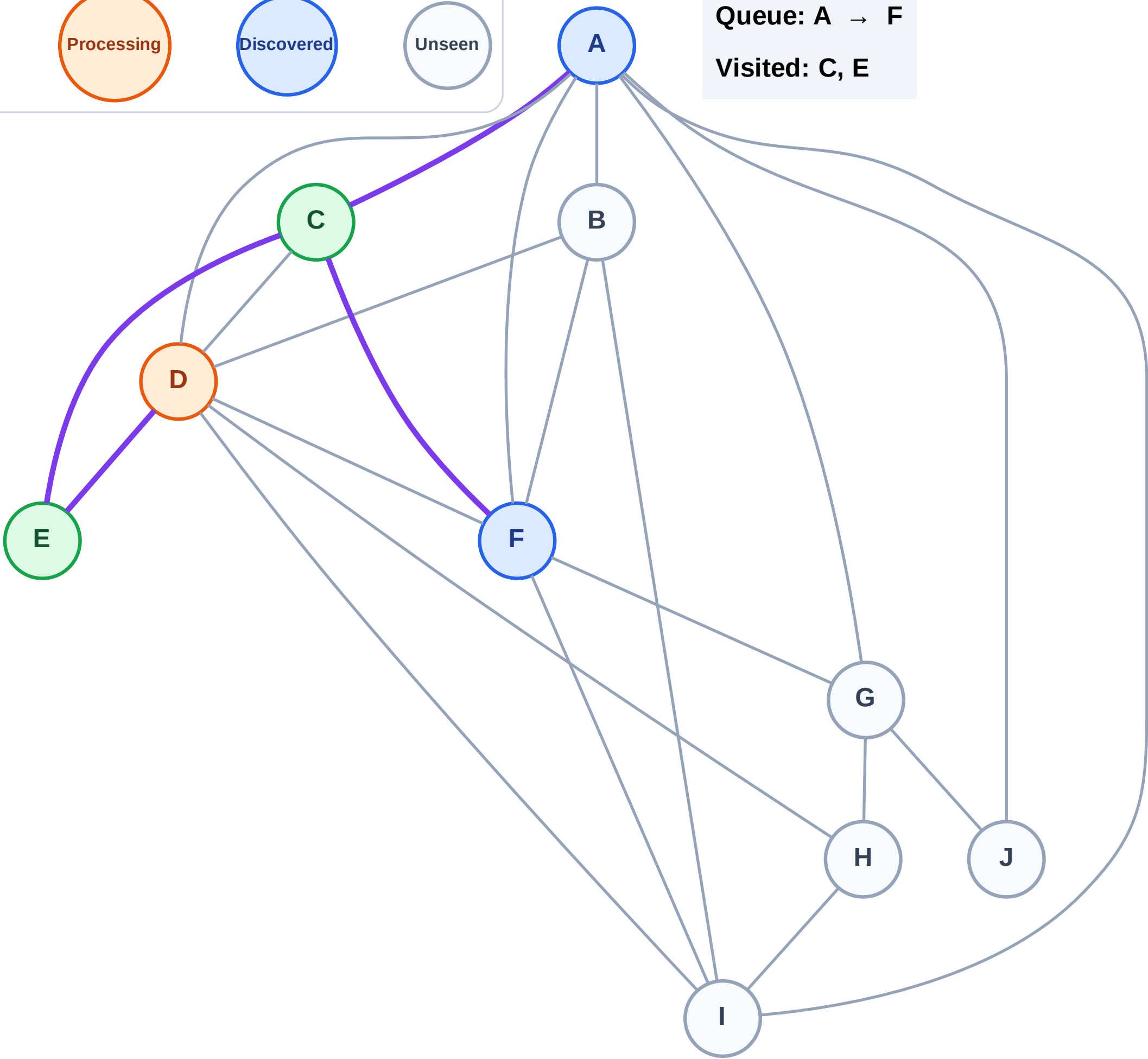
Complete

Processing

Discovered

Unseen

Queue: A → F
Visited: C, E



BFS Traversal

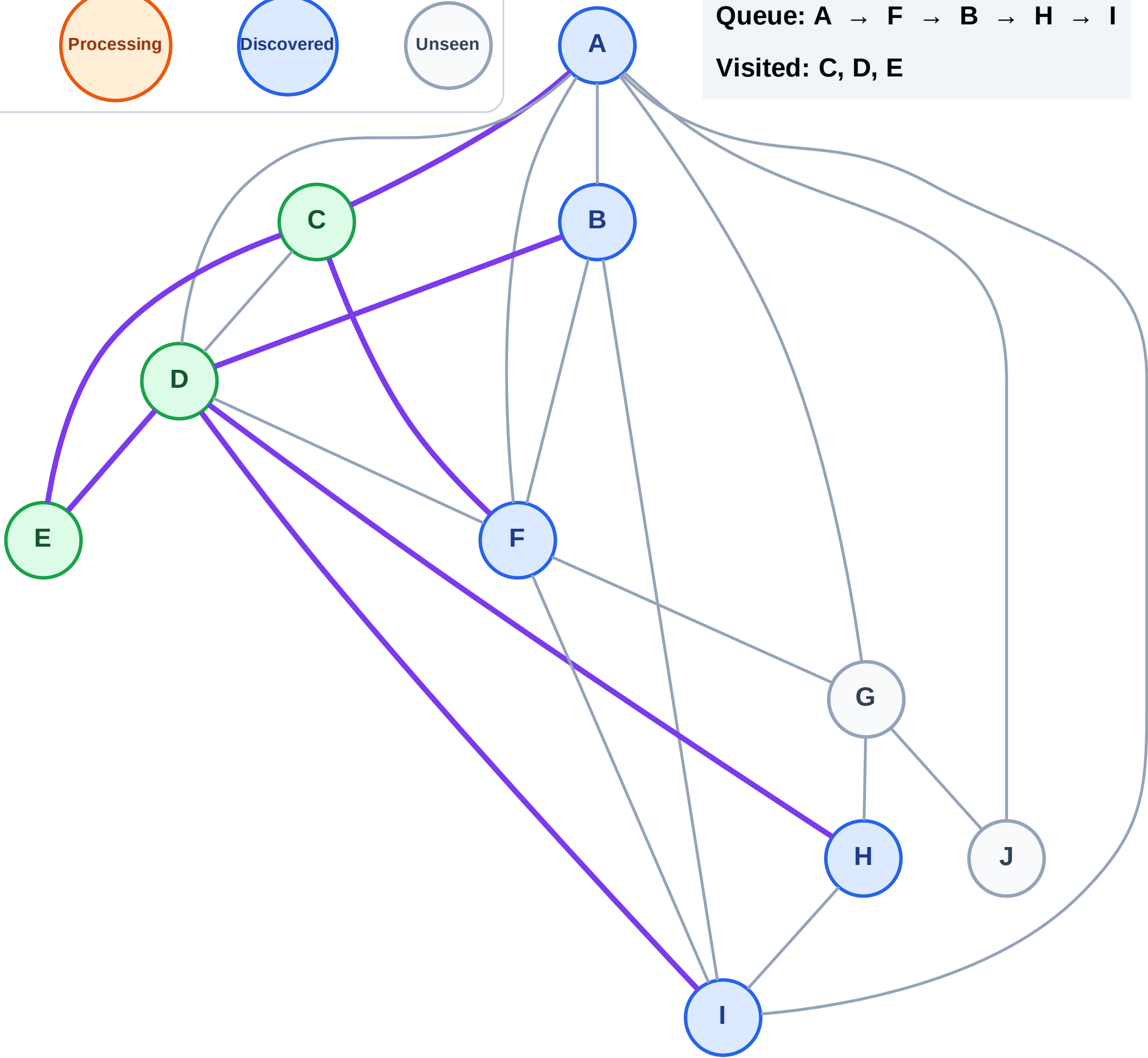
Step 7: Discover B, H, I from D

Legend



Queue: A → F → B → H → I

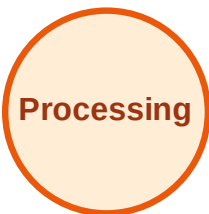
Visited: C, D, E



BFS Traversal

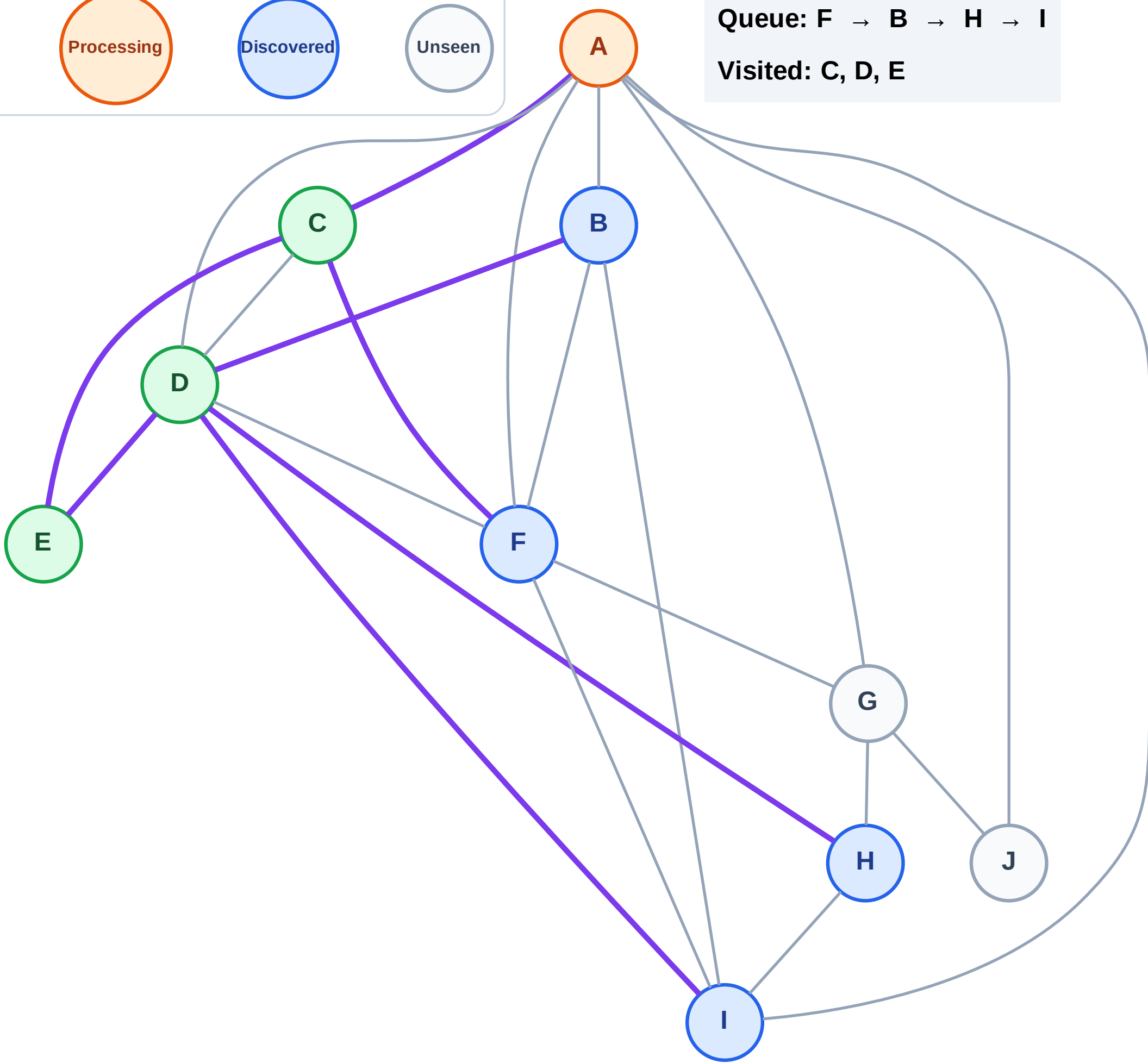
Step 8: Process node A

Legend



Queue: F → B → H → I

Visited: C, D, E



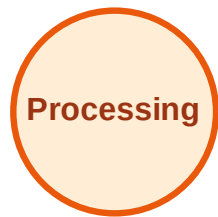
BFS Traversal

Step 9: Discover G, J from A

Legend



Complete



Processing



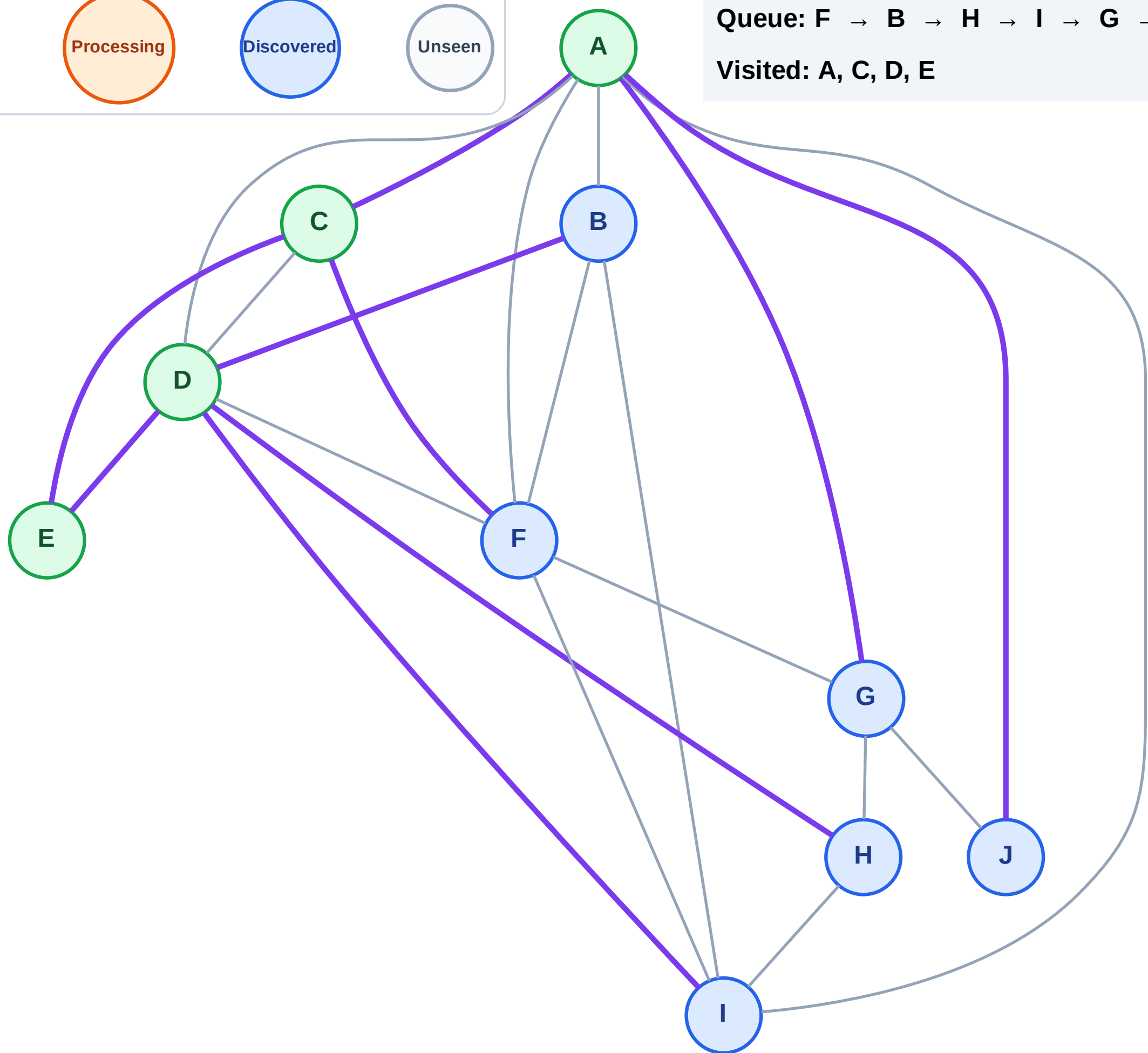
Discovered



Unseen

Queue: F → B → H → I → G → J

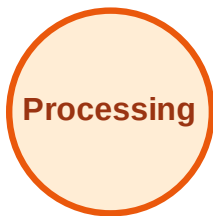
Visited: A, C, D, E



BFS Traversal

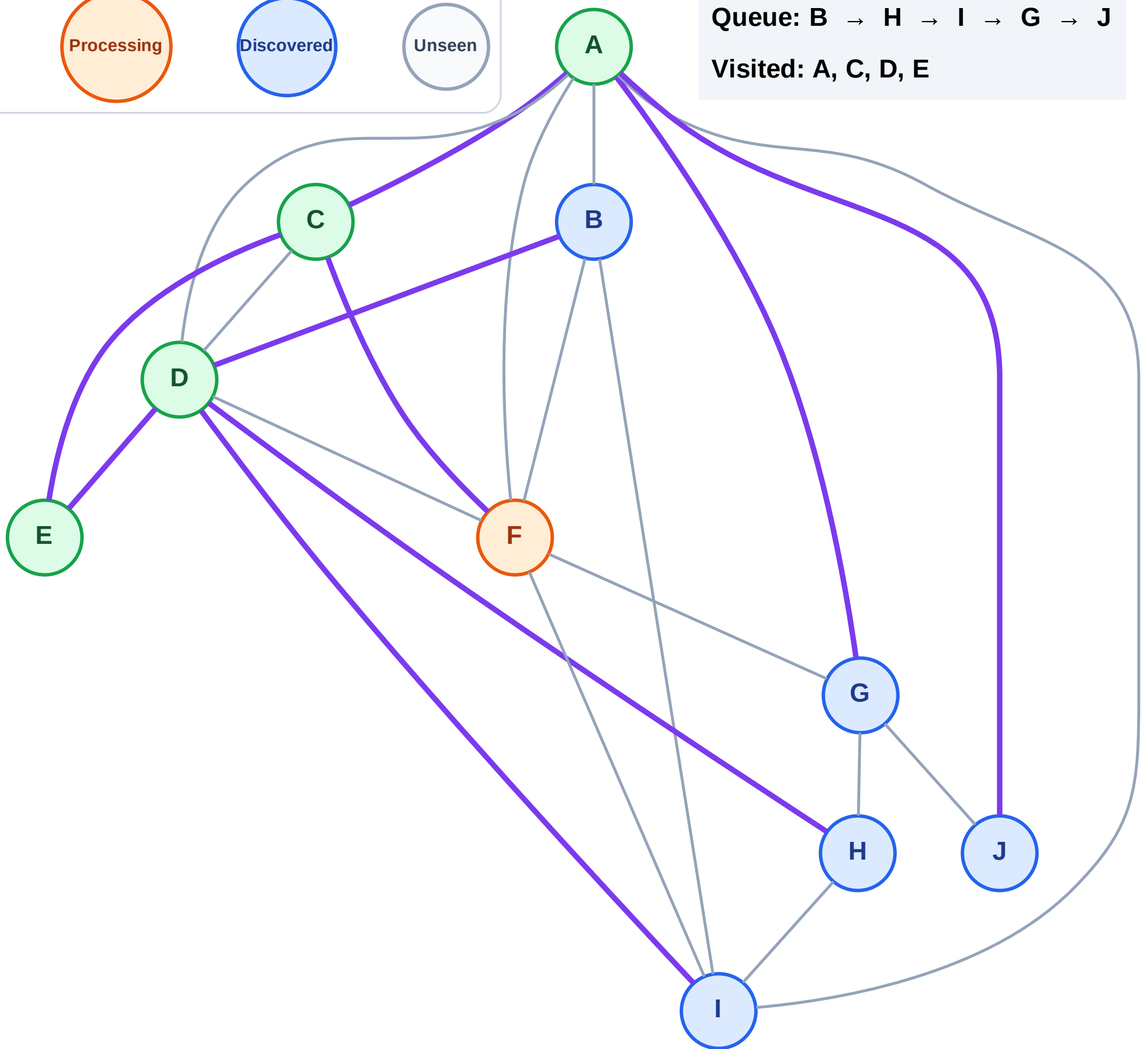
Step 10: Process node F

Legend



Queue: B → H → I → G → J

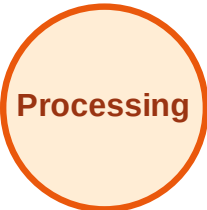
Visited: A, C, D, E



BFS Traversal

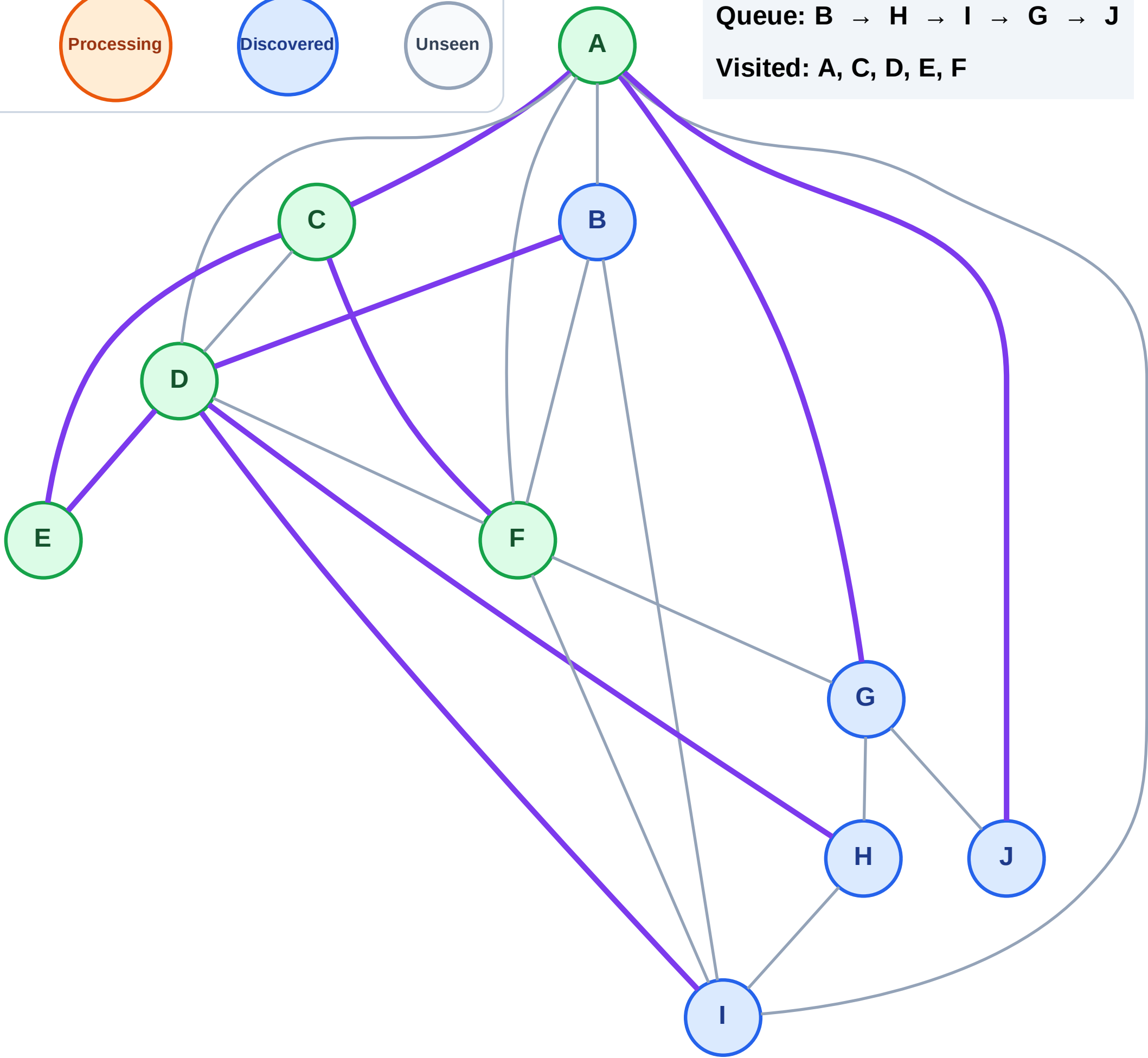
Step 11: No new neighbors from F

Legend



Queue: B → H → I → G → J

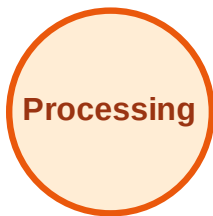
Visited: A, C, D, E, F



BFS Traversal

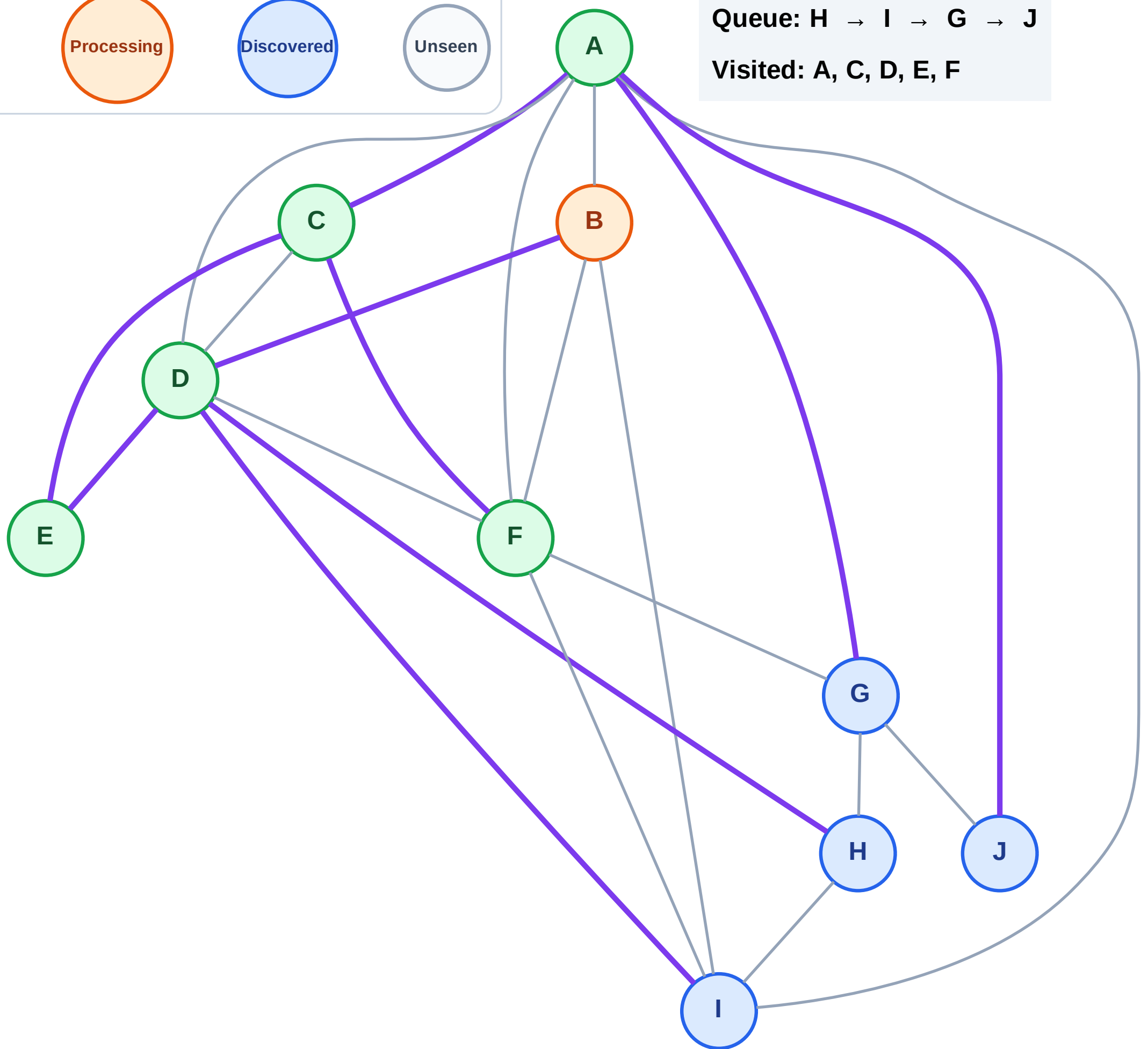
Step 12: Process node B

Legend



Queue: H → I → G → J

Visited: A, C, D, E, F



BFS Traversal

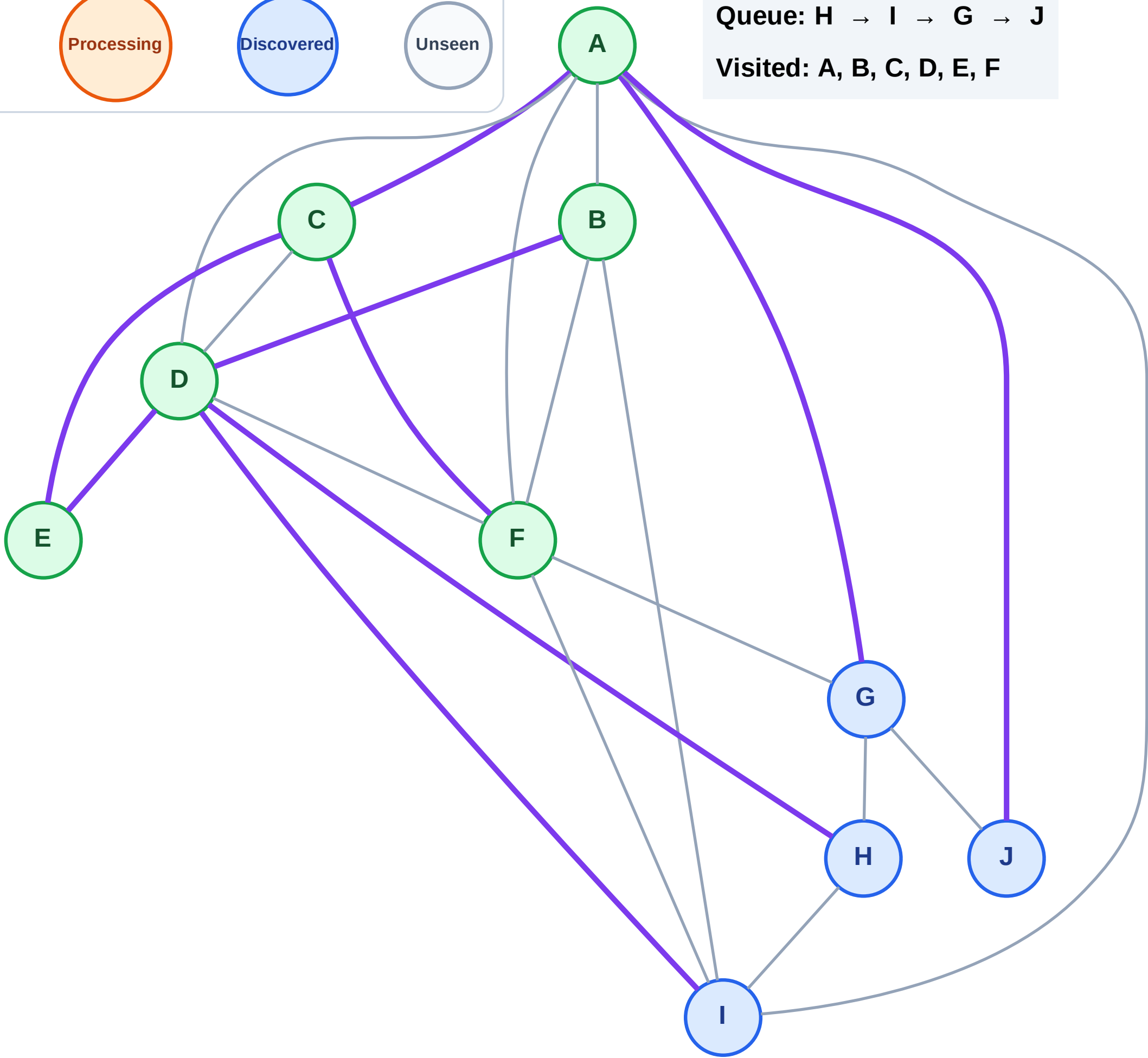
Step 13: No new neighbors from B

Legend



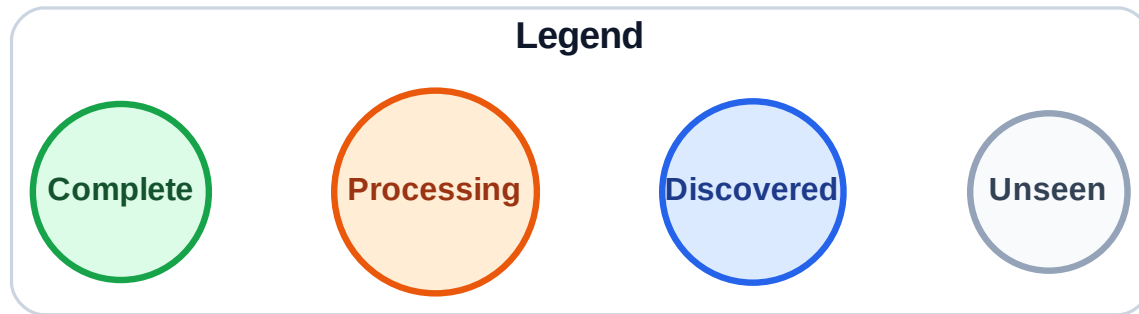
Queue: H → I → G → J

Visited: A, B, C, D, E, F

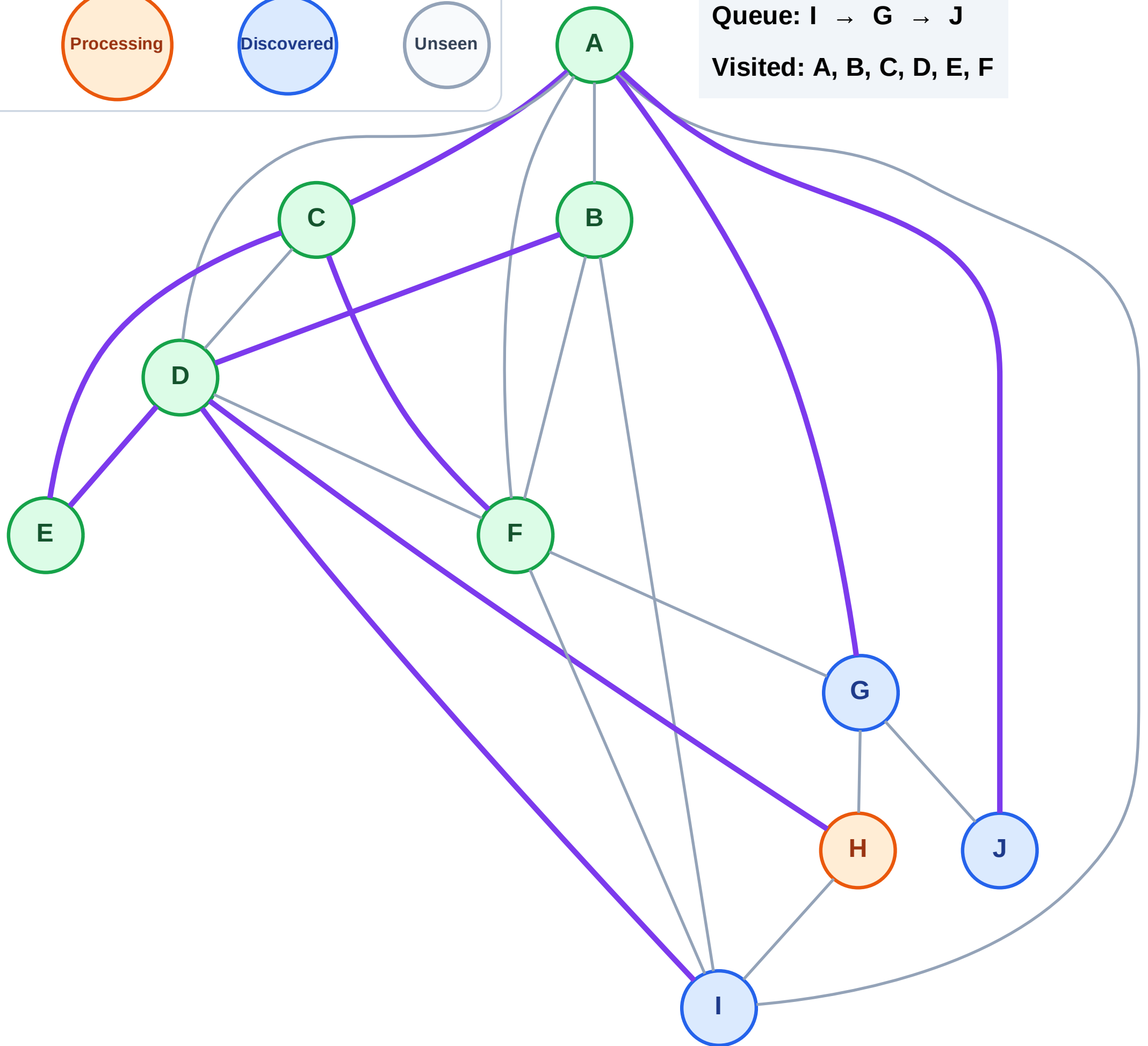


BFS Traversal

Step 14: Process node H



Queue: I → G → J
Visited: A, B, C, D, E, F



BFS Traversal

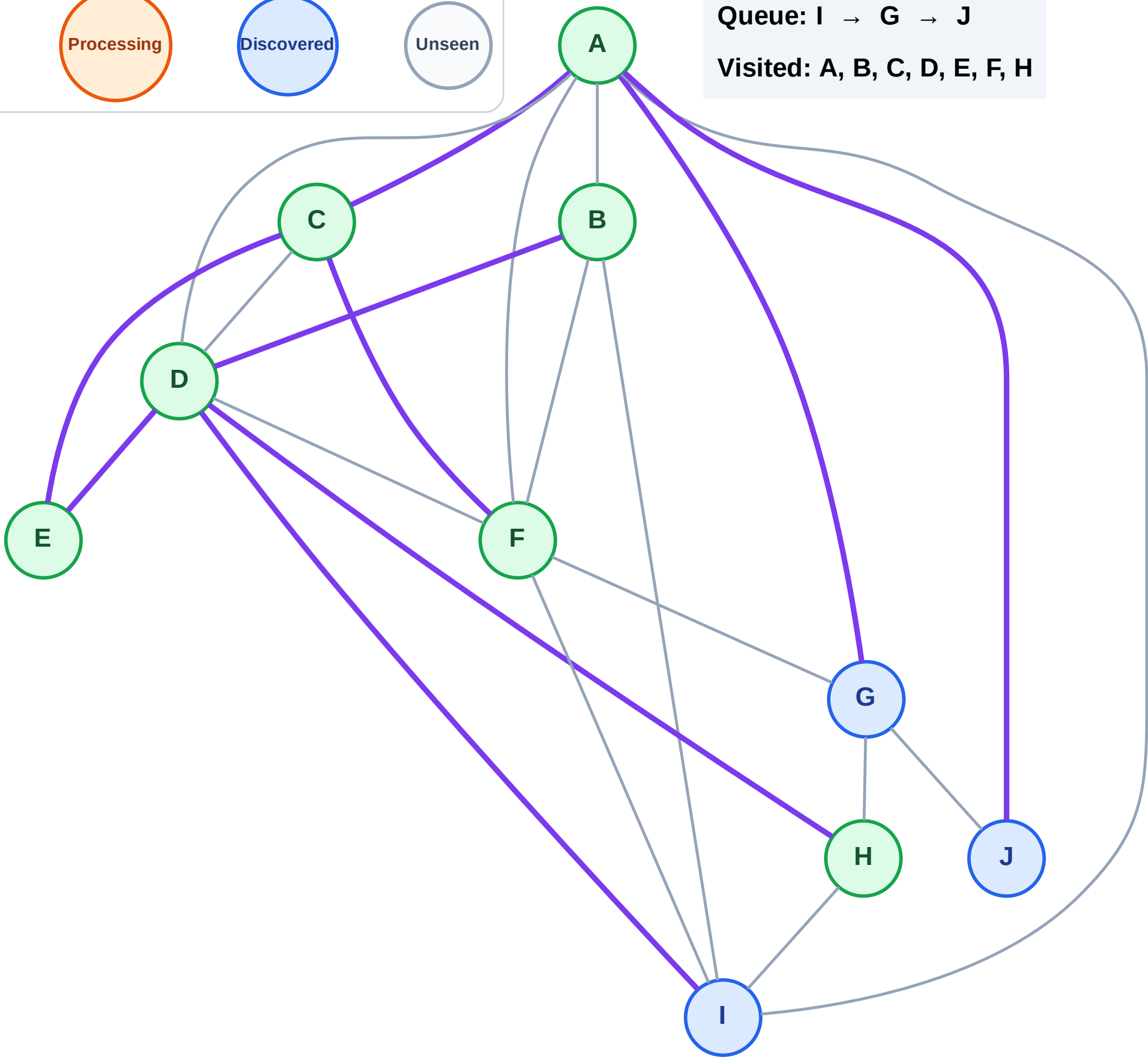
Step 15: No new neighbors from H

Legend



Queue: I → G → J

Visited: A, B, C, D, E, F, H



BFS Traversal

Step 16: Process node I

Legend

Complete

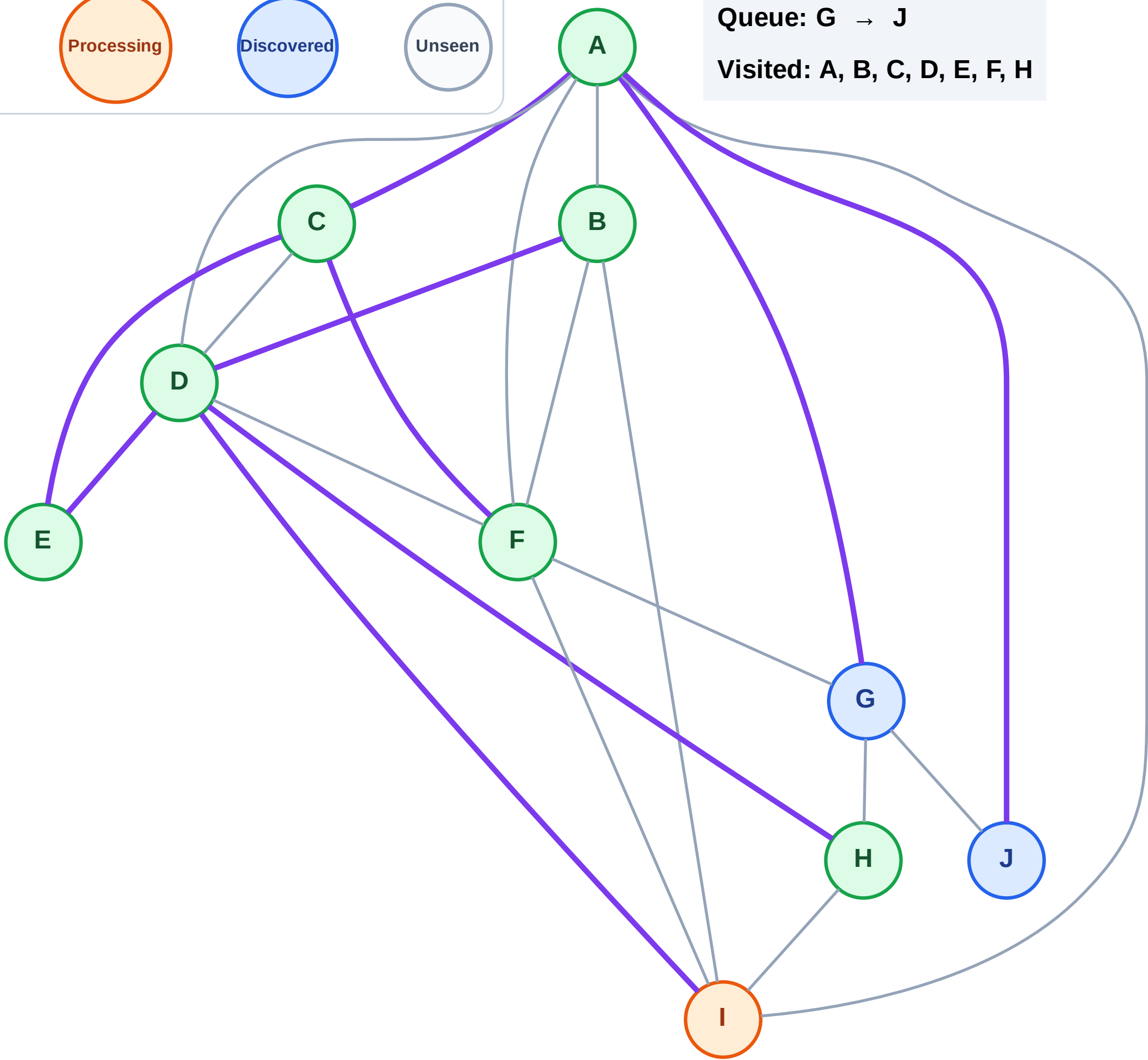
Processing

Discovered

Unseen

Queue: G → J

Visited: A, B, C, D, E, F, H



BFS Traversal

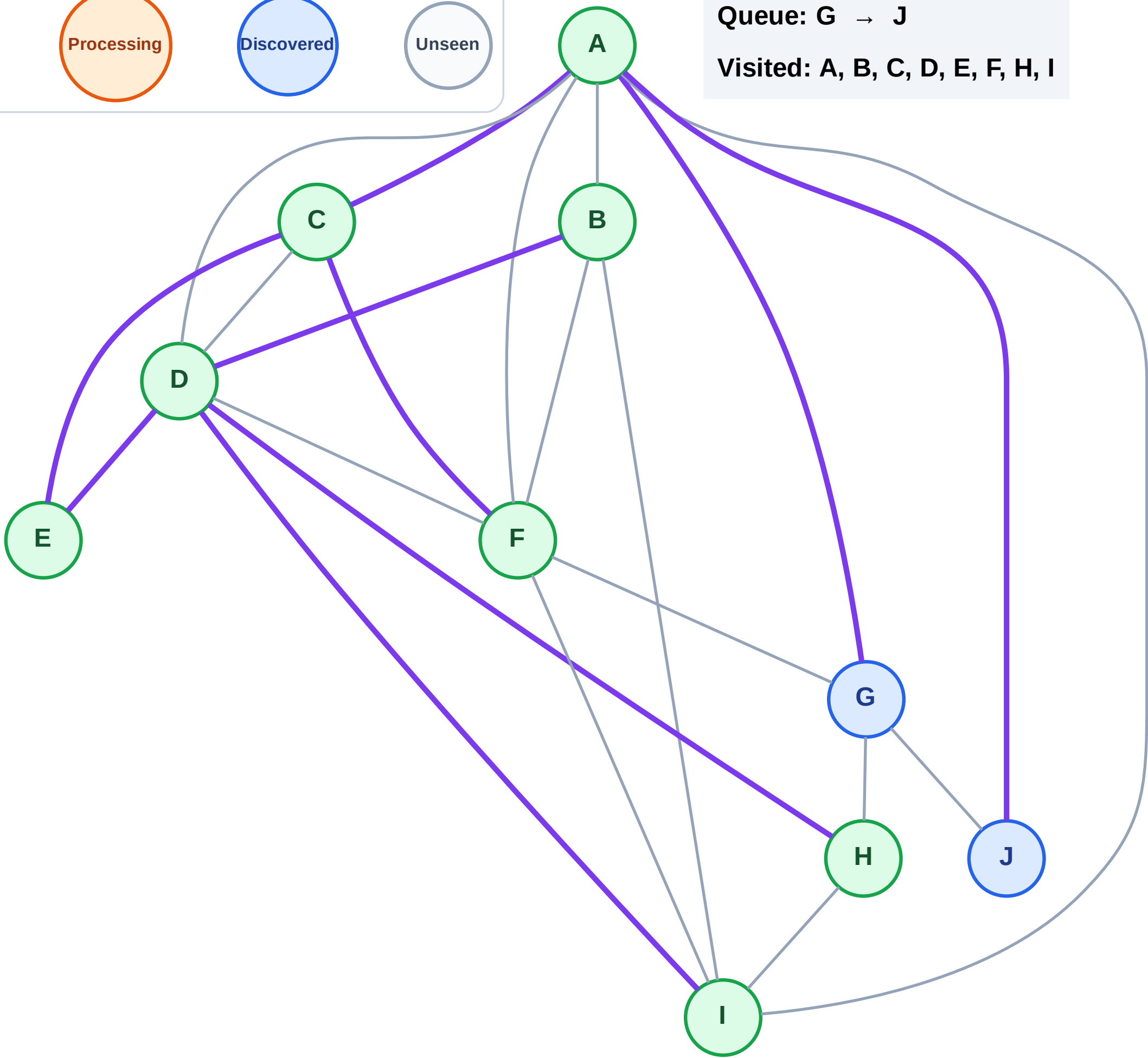
Step 17: No new neighbors from I

Legend



Queue: G → J

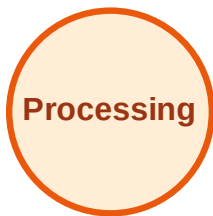
Visited: A, B, C, D, E, F, H, I



BFS Traversal

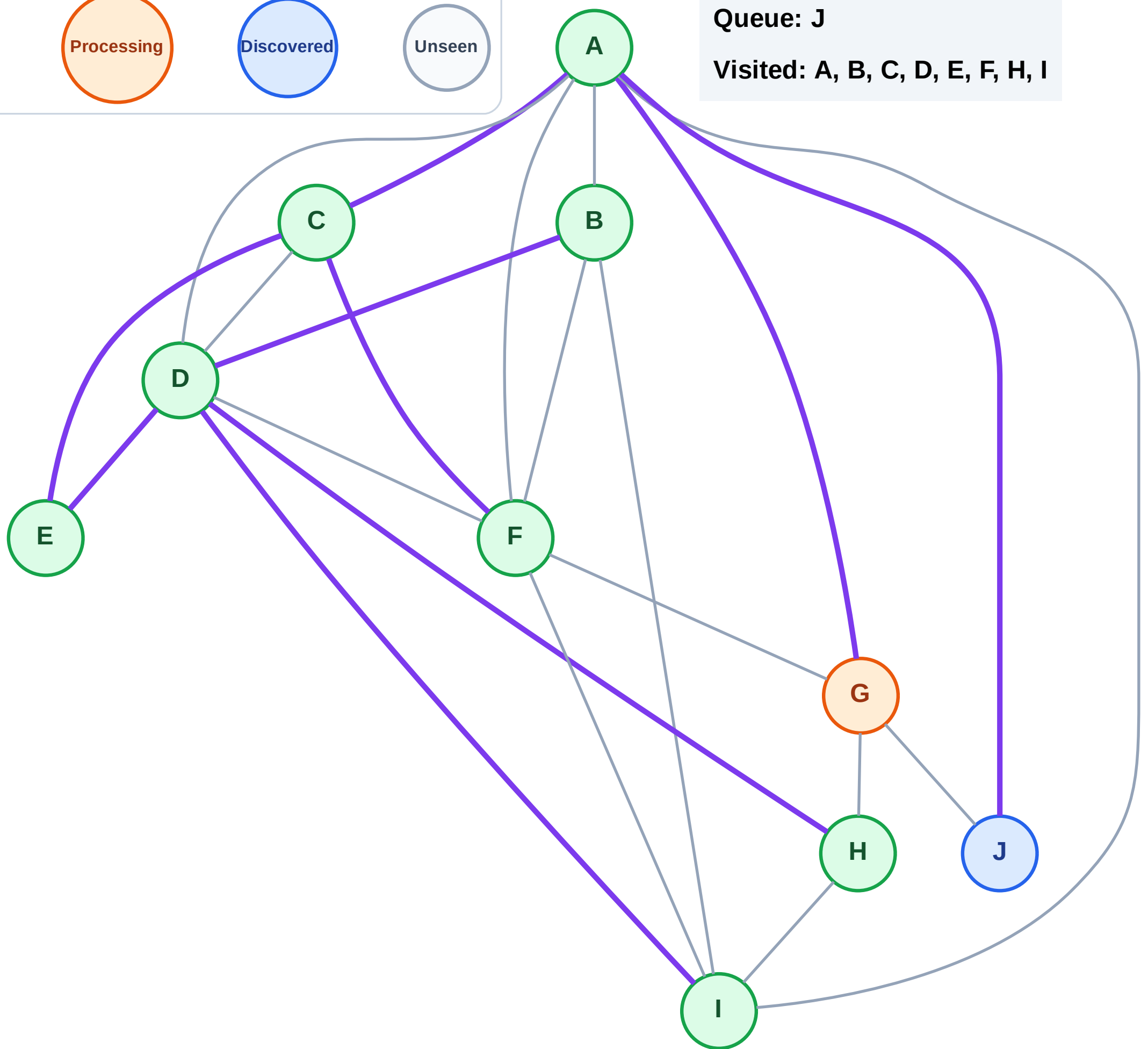
Step 18: Process node G

Legend



Queue: J

Visited: A, B, C, D, E, F, H, I



BFS Traversal

Step 19: No new neighbors from G

Legend

Complete

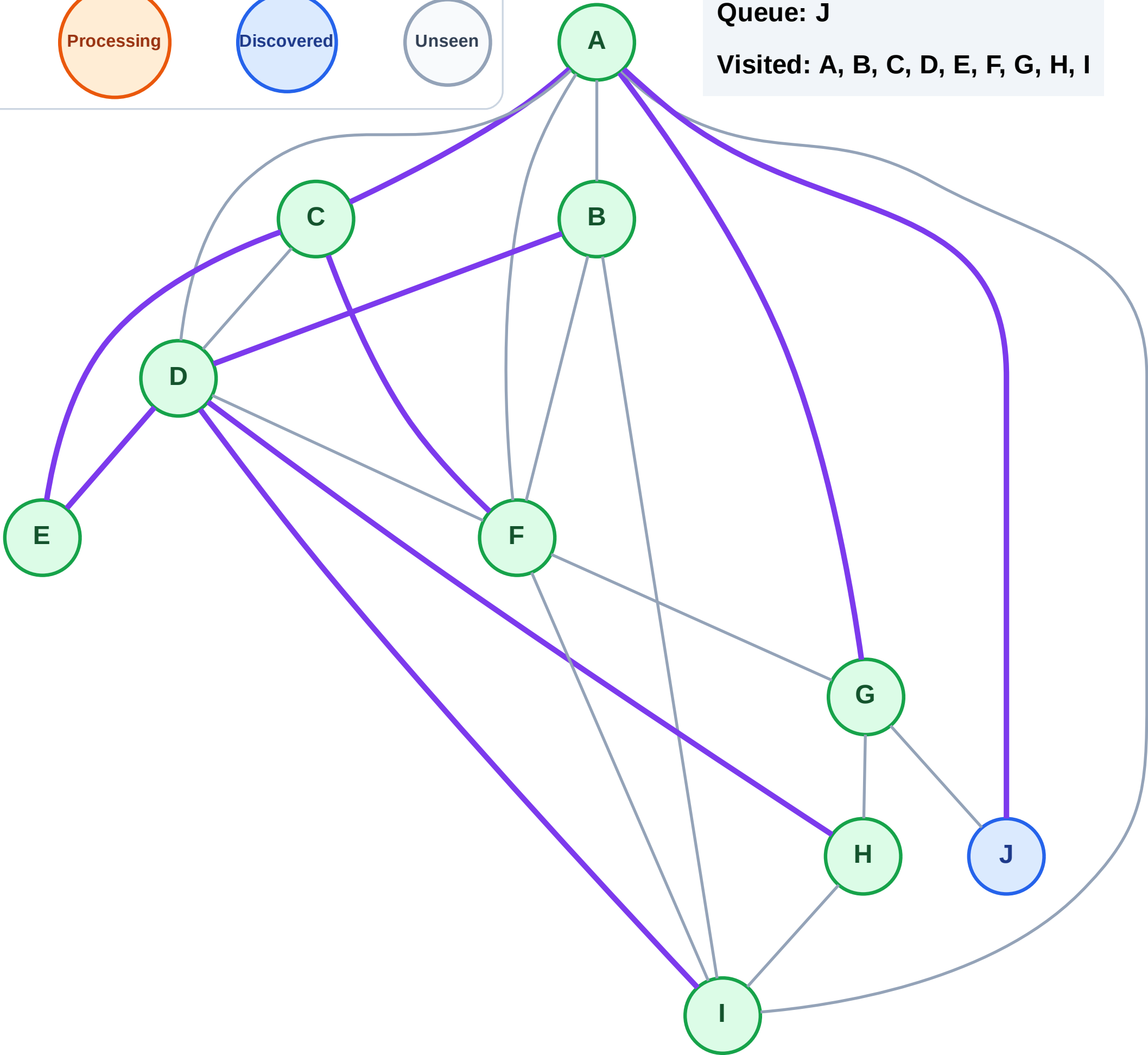
Processing

Discovered

Unseen

Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

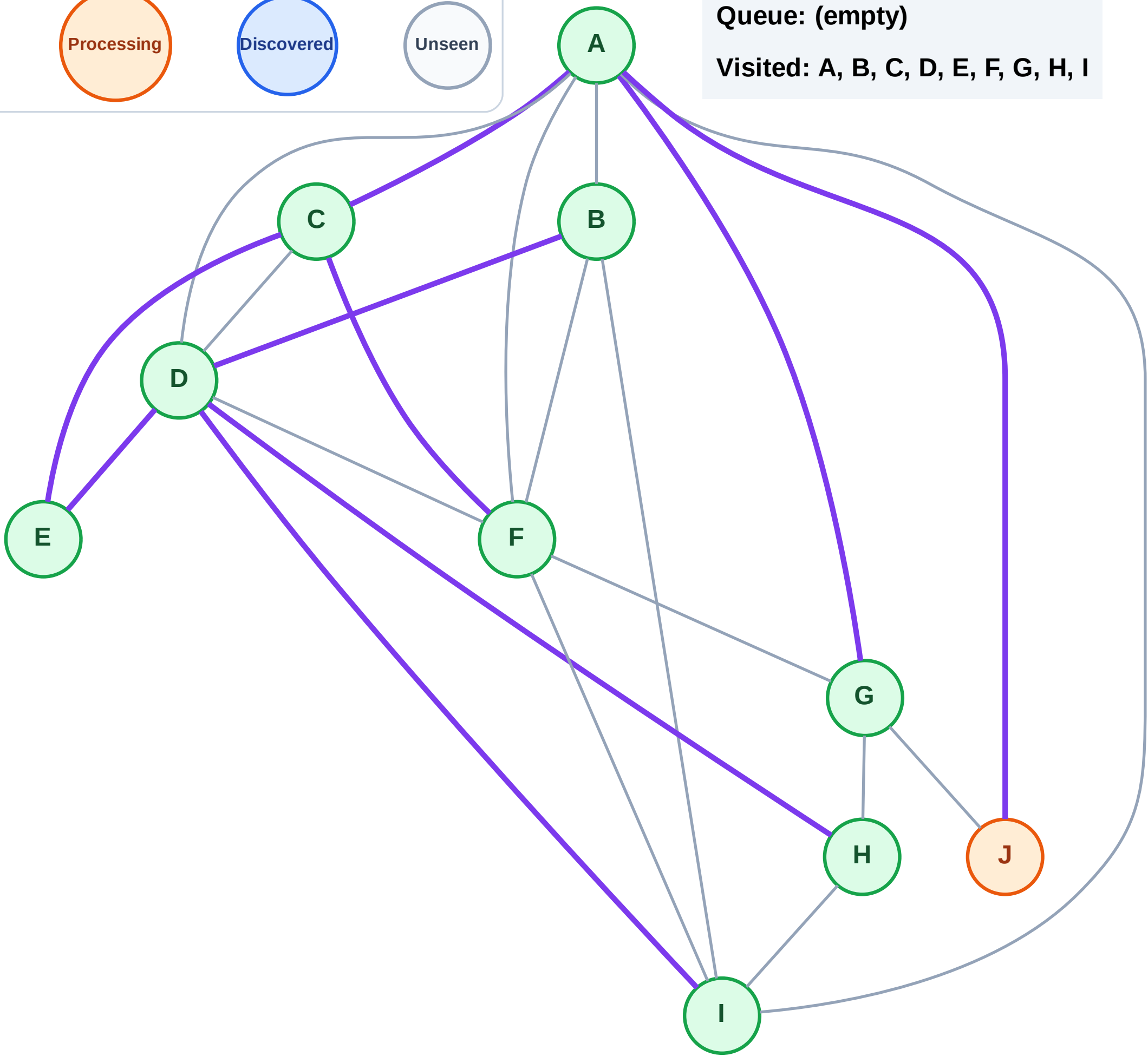
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

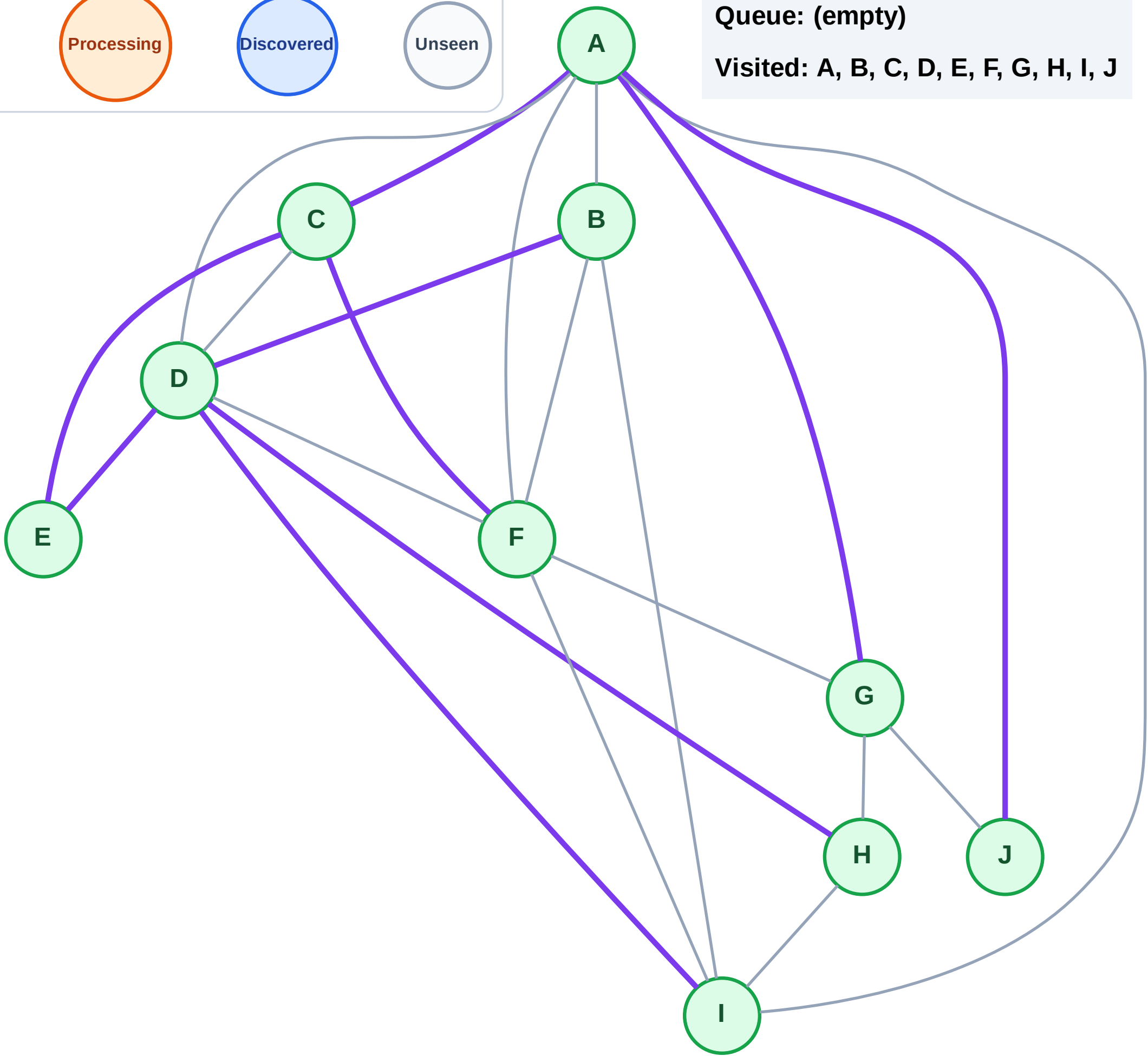
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

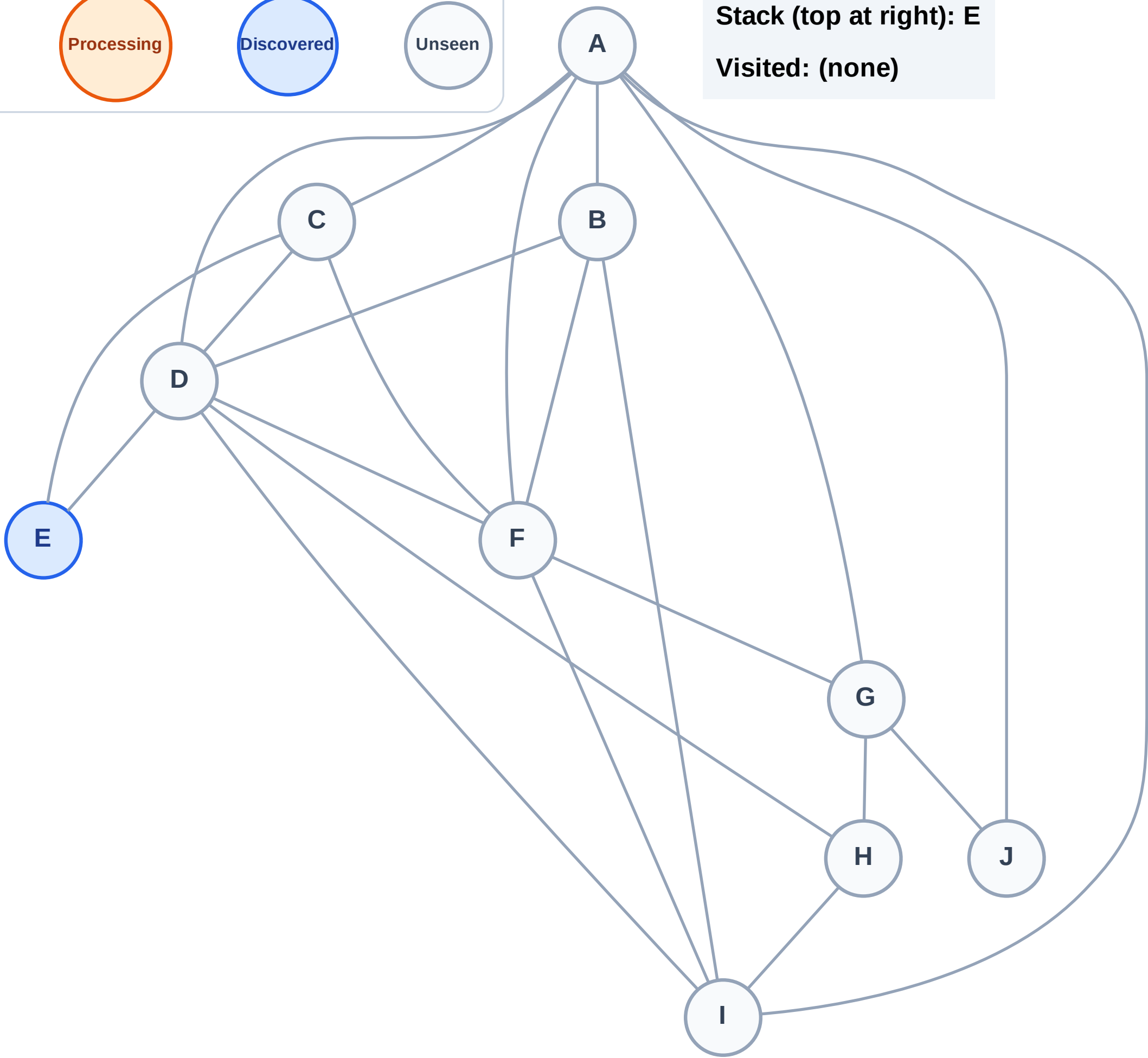
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

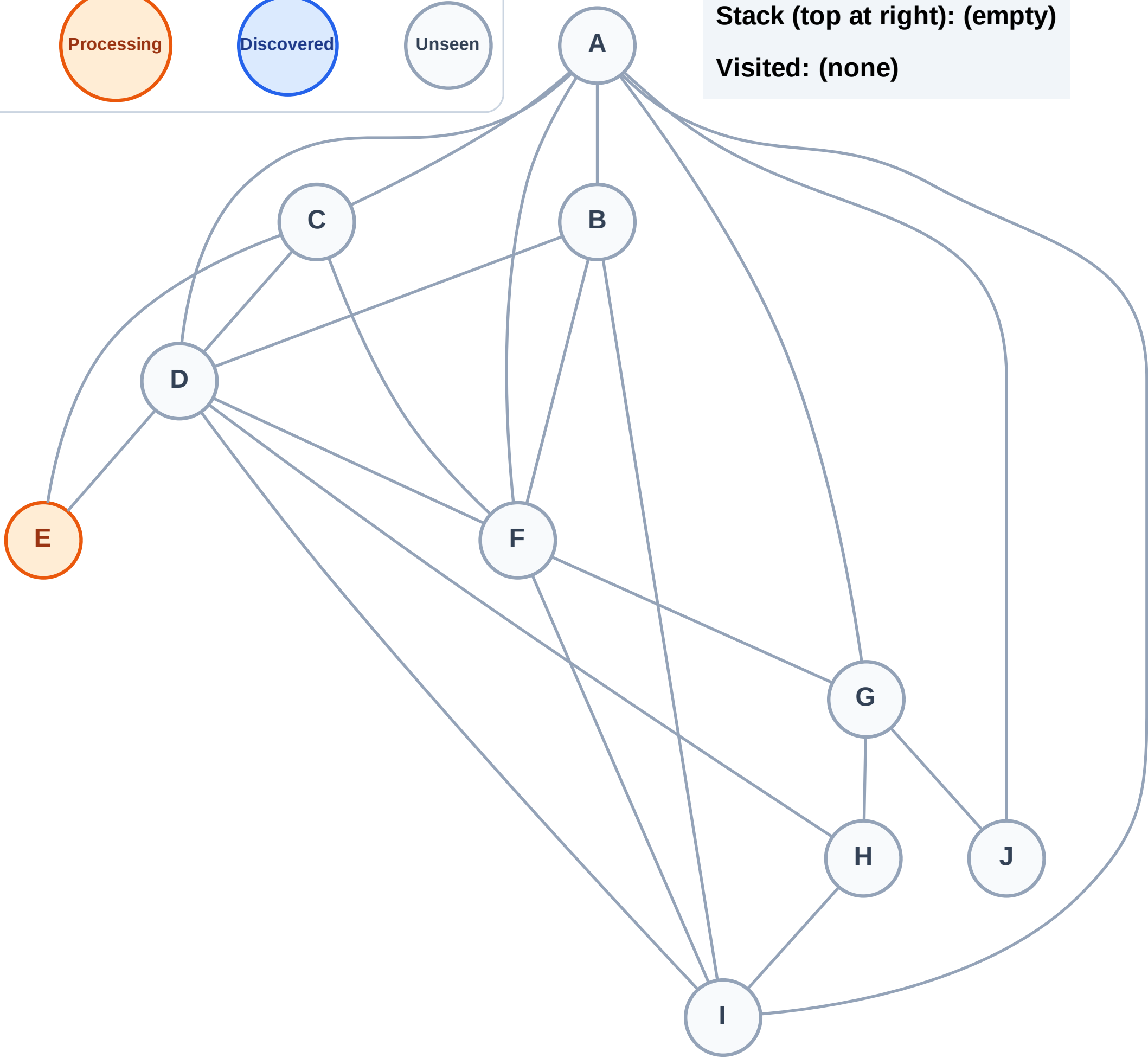
Step 2: Process node E

Legend



Stack (top at right): (empty)

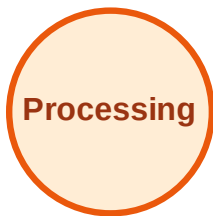
Visited: (none)



DFS Traversal

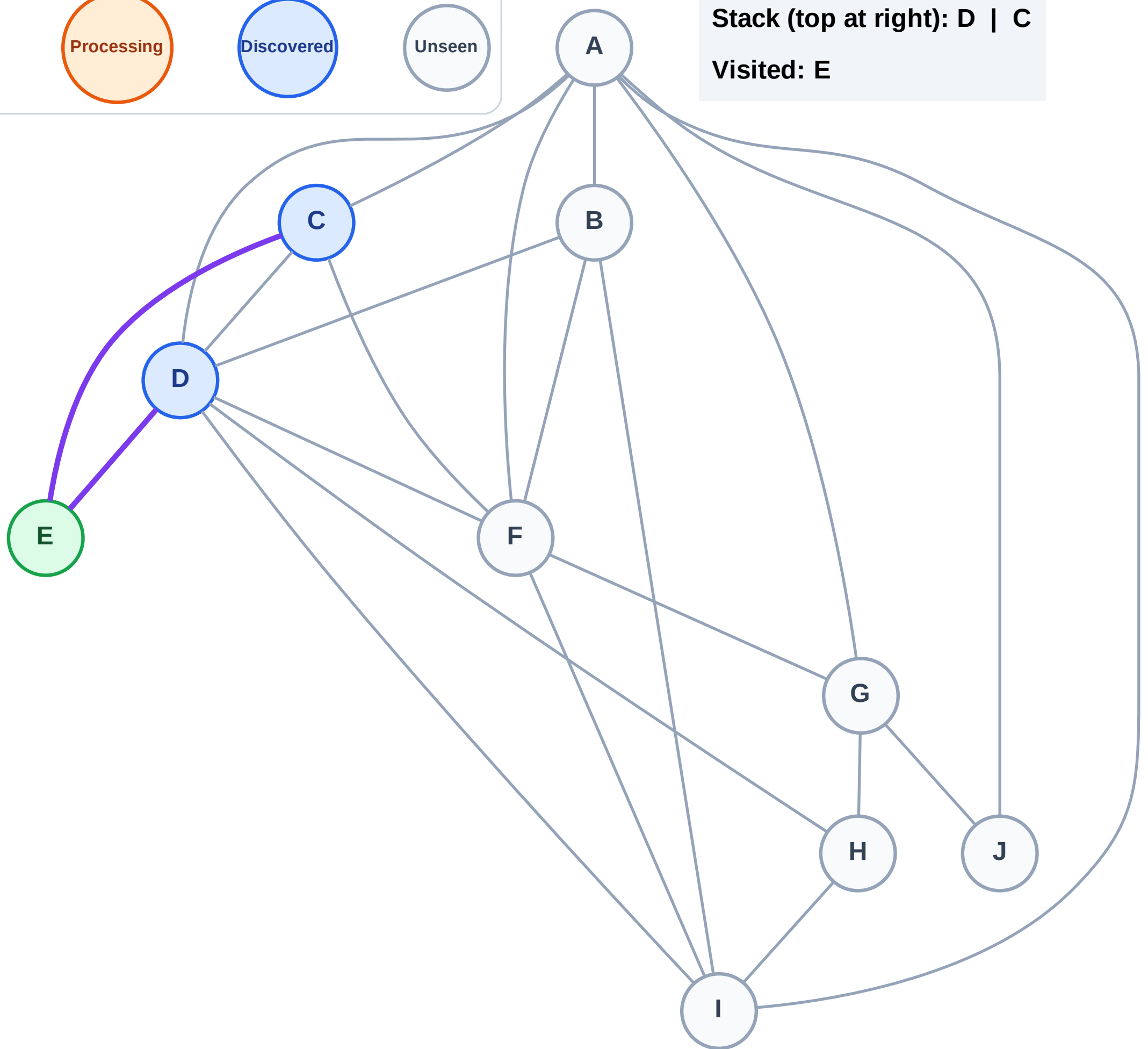
Step 3: Discover D, C from E

Legend



Stack (top at right): D | C

Visited: E



DFS Traversal

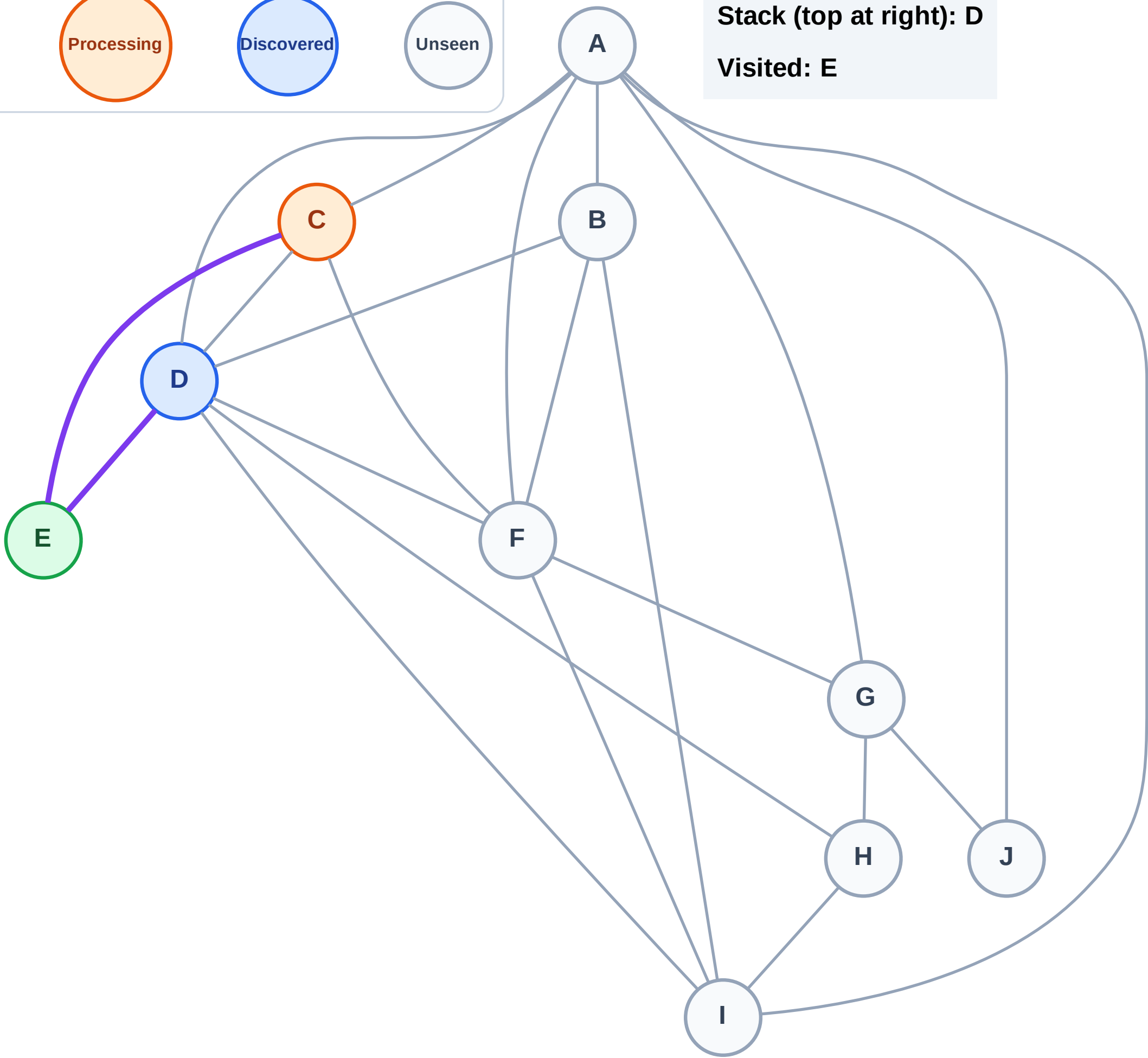
Step 4: Process node C

Legend



Stack (top at right): D

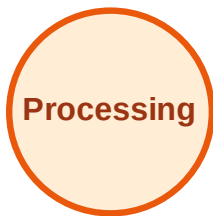
Visited: E



DFS Traversal

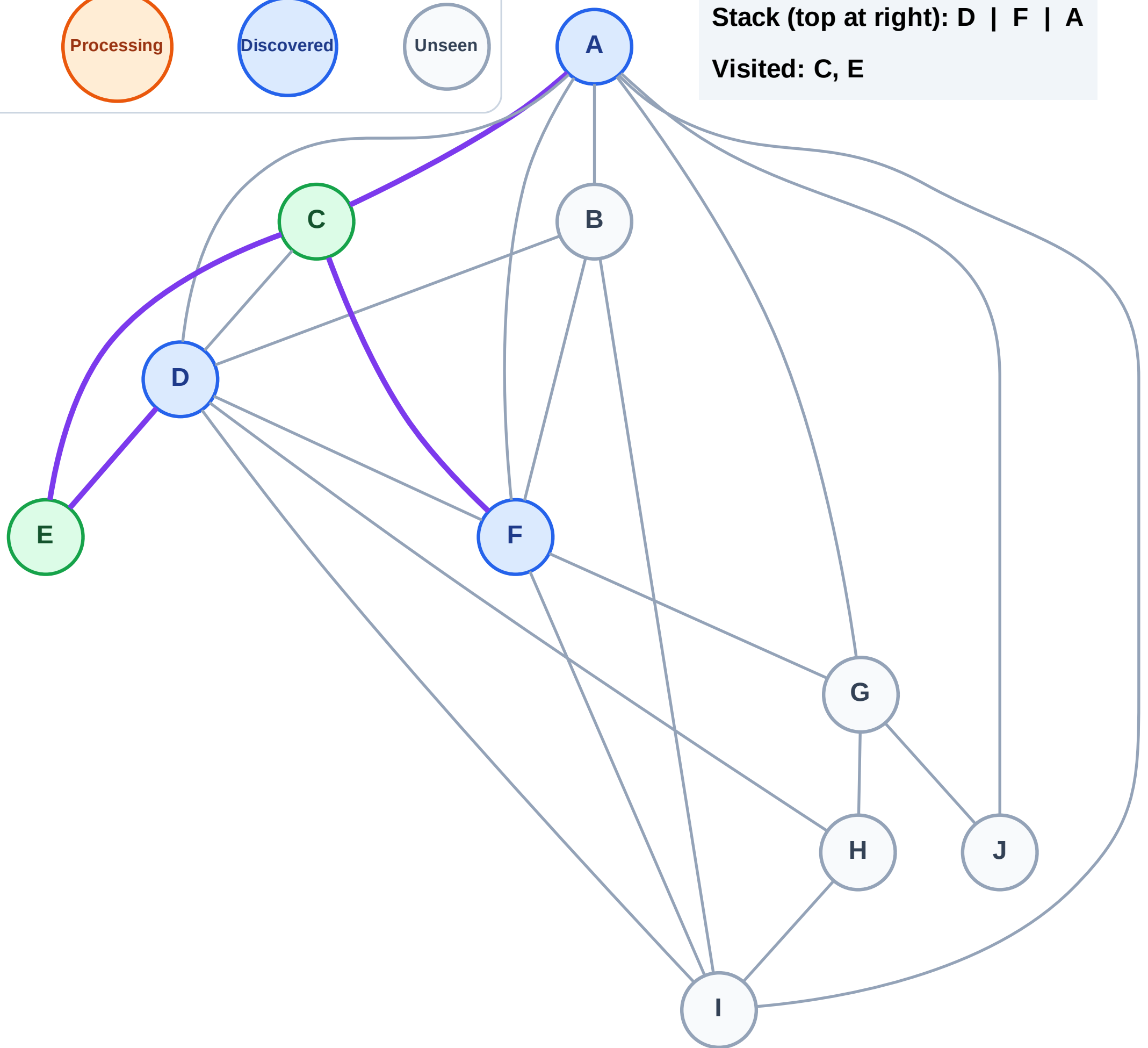
Step 5: Discover F, A from C

Legend



Stack (top at right): D | F | A

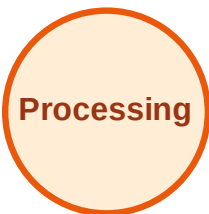
Visited: C, E



DFS Traversal

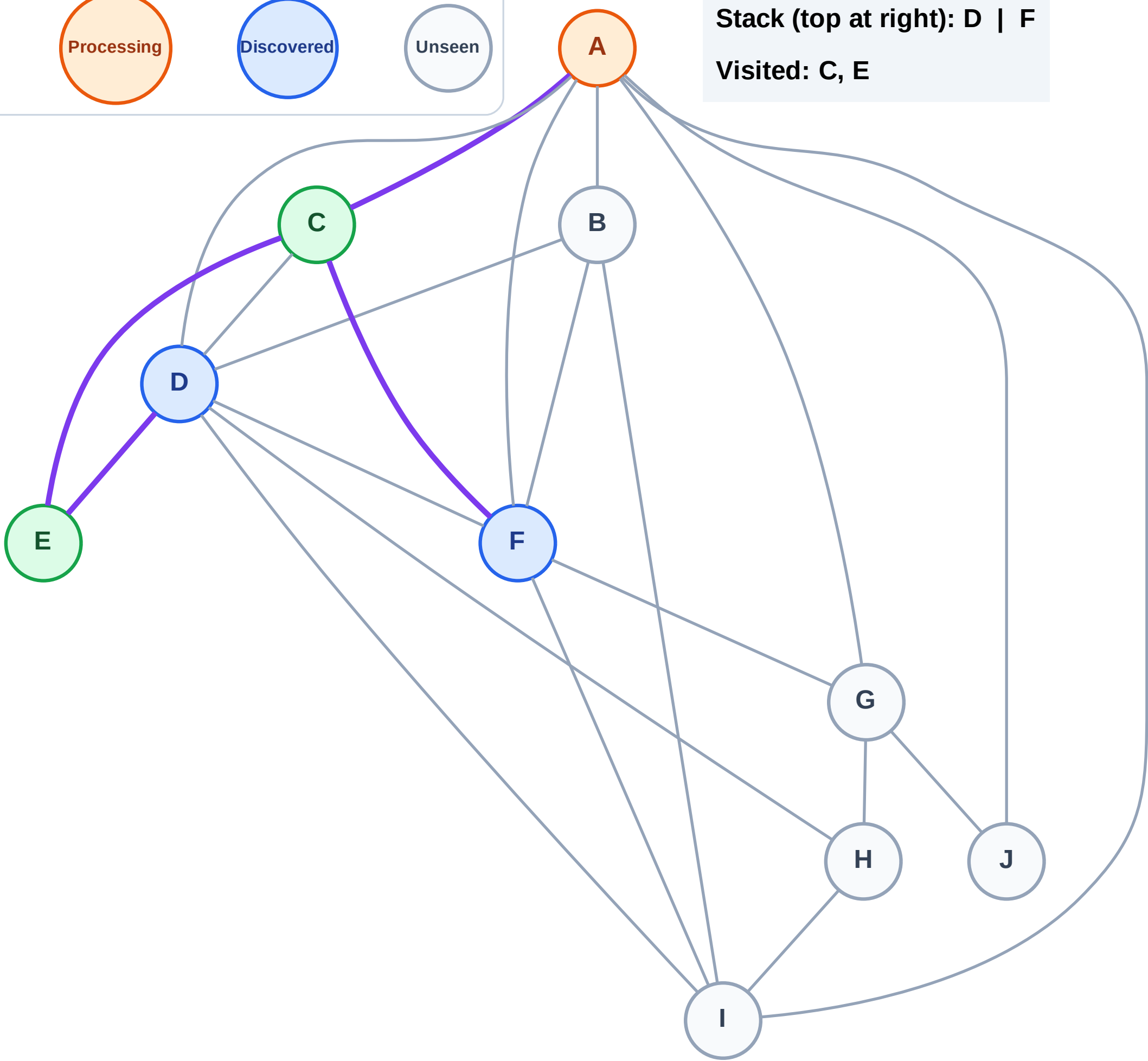
Step 6: Process node A

Legend



Stack (top at right): D | F

Visited: C, E



DFS Traversal

Step 7: Discover J, I, G, B from A

Legend



Complete



Processing



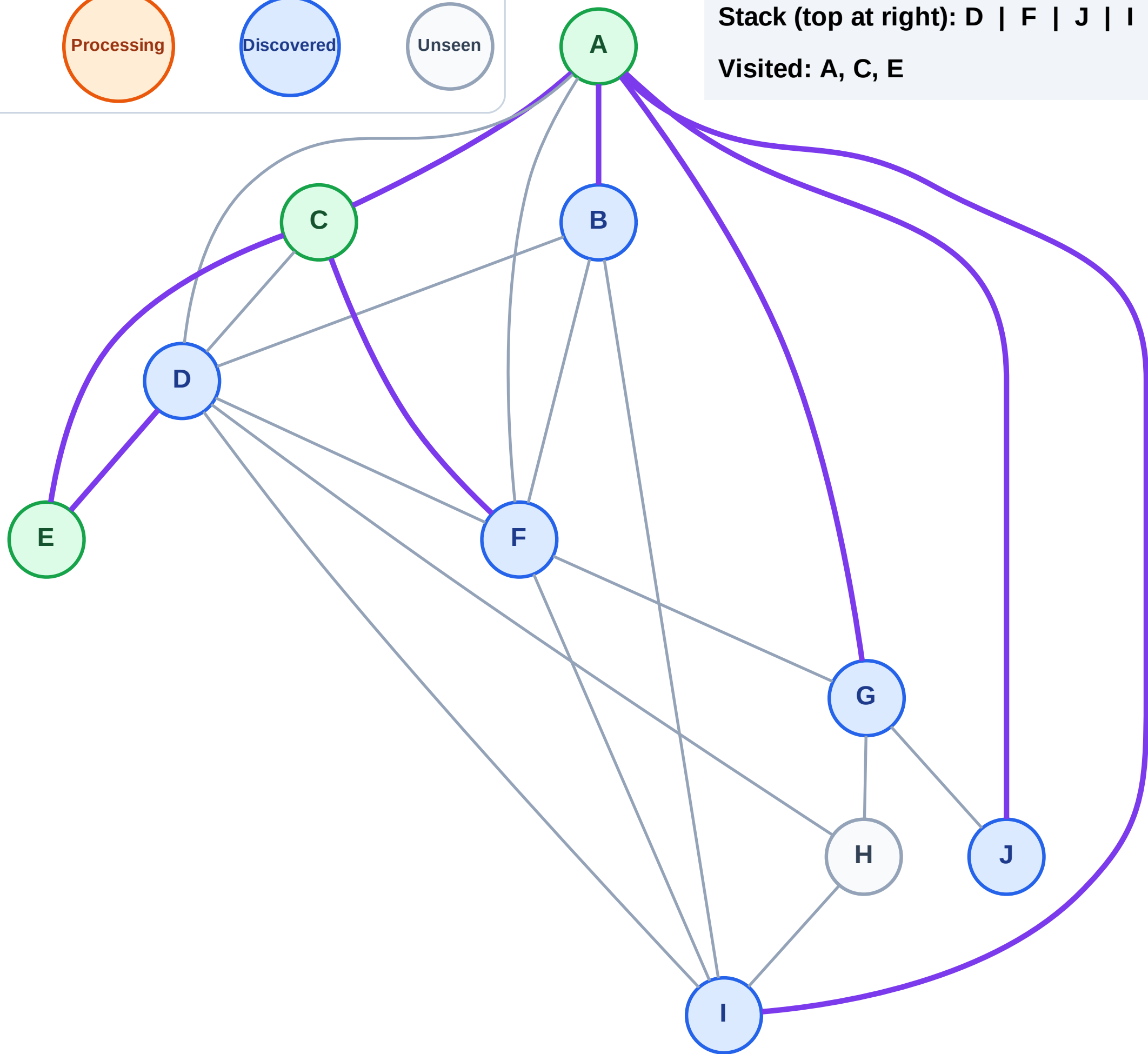
Discovered



Unseen

Stack (top at right): D | F | J | I | G | B

Visited: A, C, E

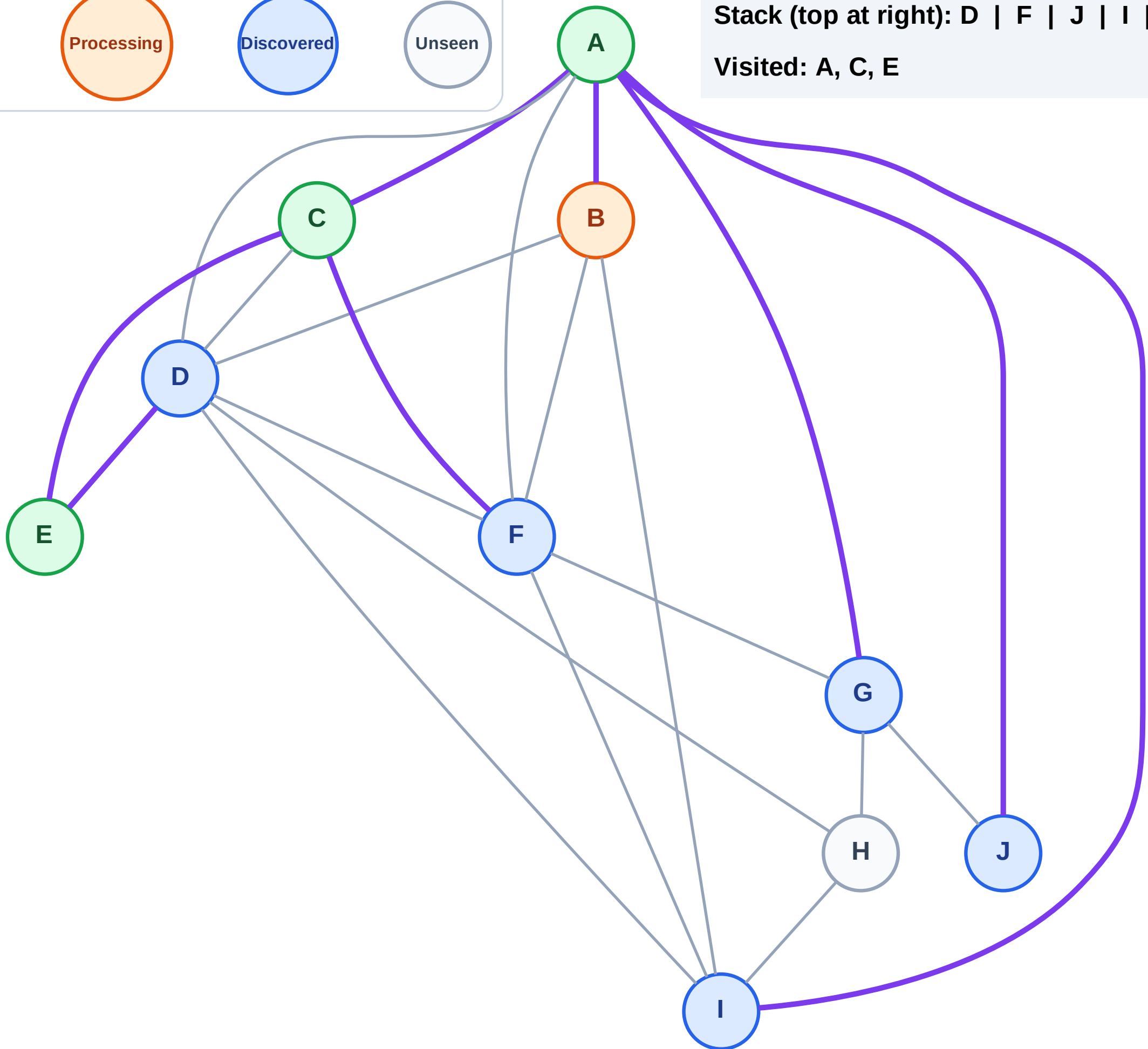
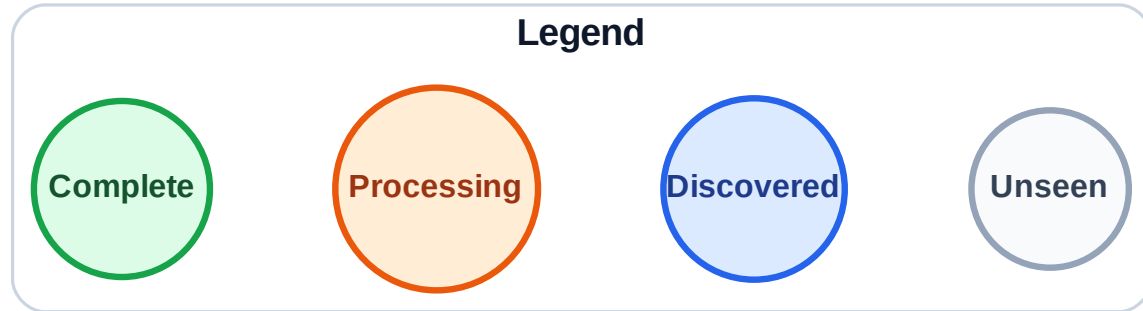


DFS Traversal

Step 8: Process node B

Stack (top at right): D | F | J | I | G

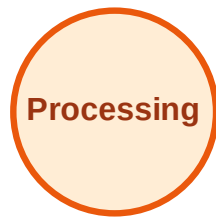
Visited: A, C, E



DFS Traversal

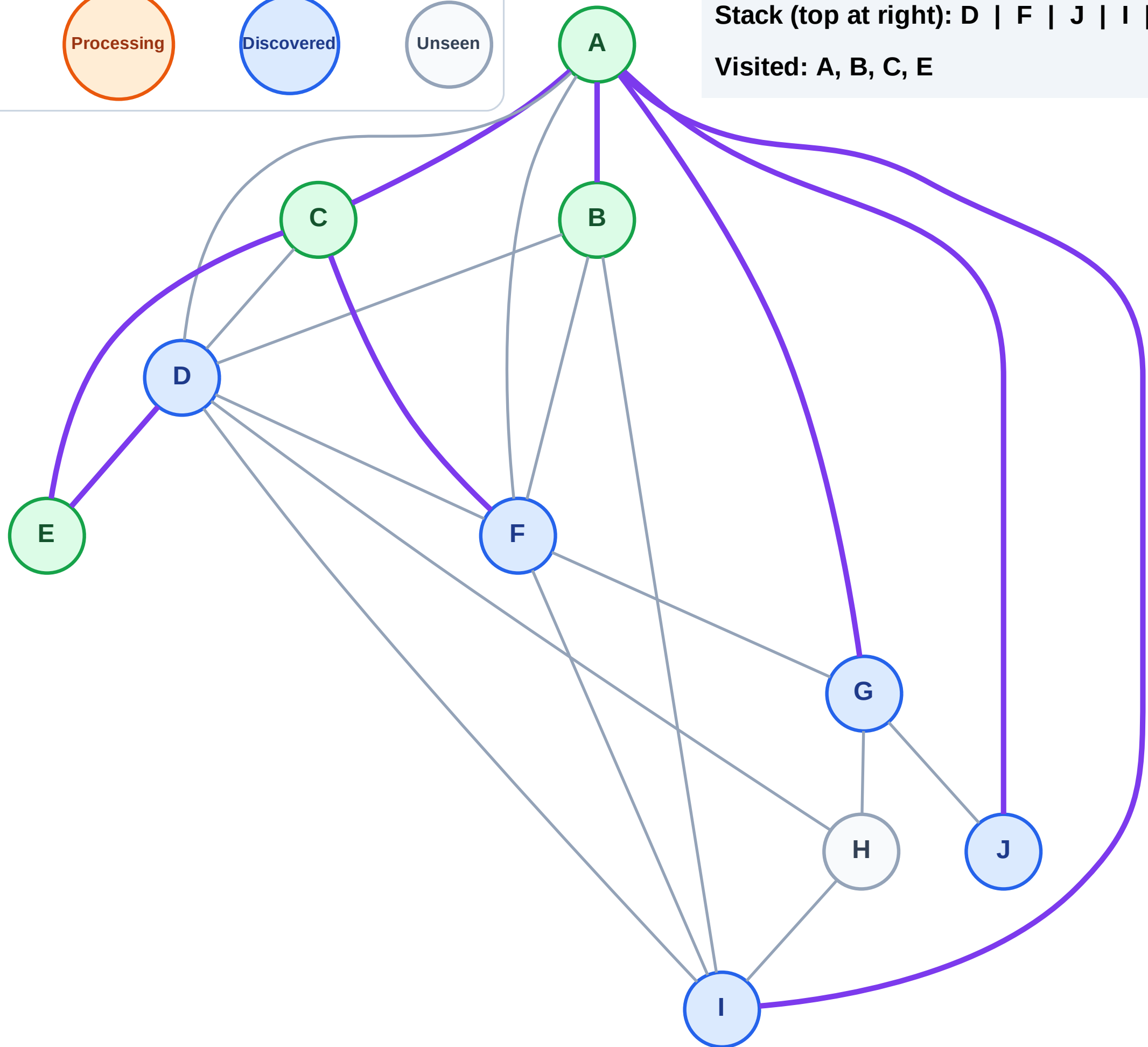
Step 9: No new neighbors from B

Legend



Stack (top at right): D | F | J | I | G

Visited: A, B, C, E



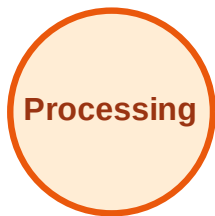
DFS Traversal

Step 10: Process node G

Legend



Complete



Processing



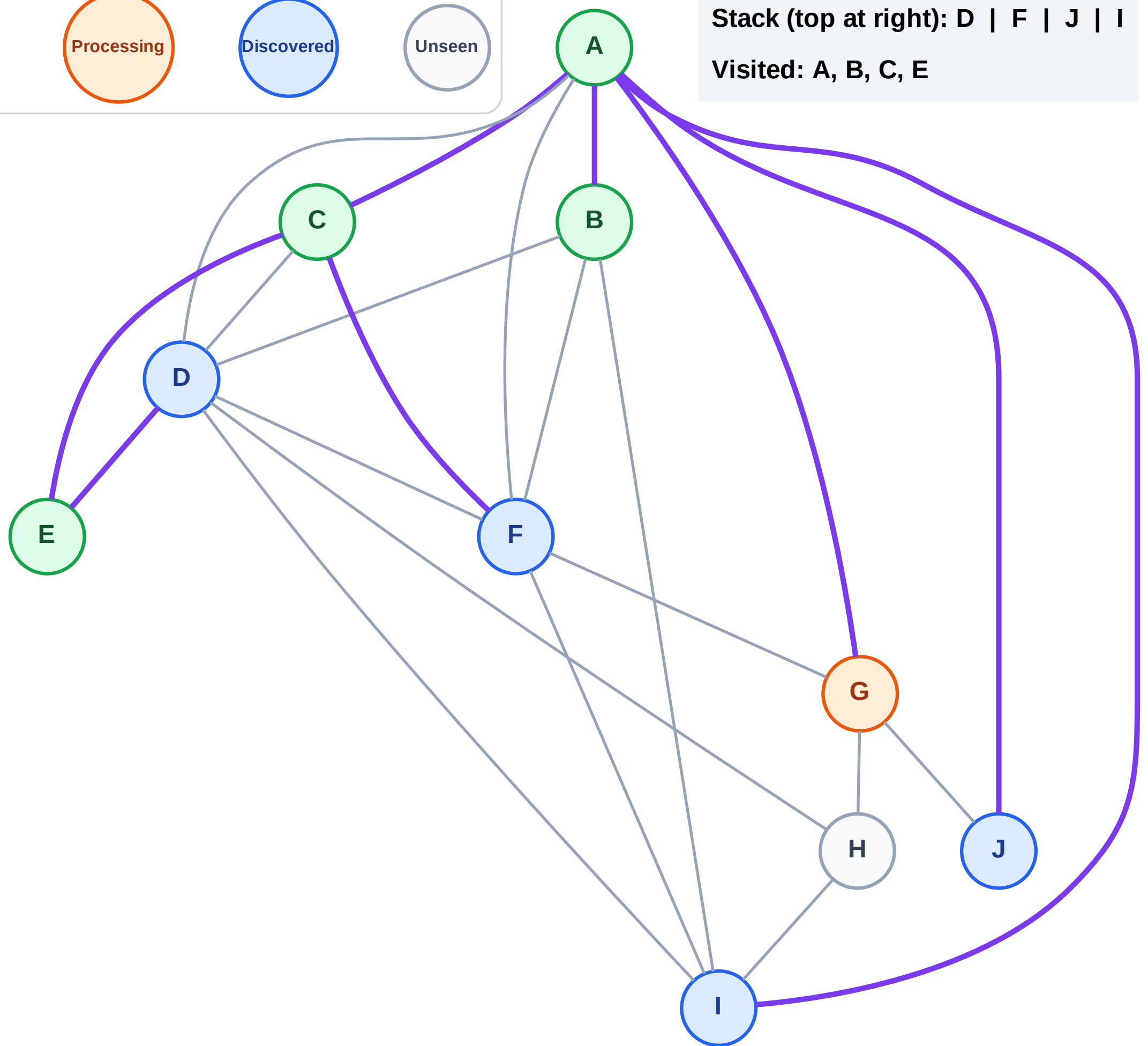
Discovered



Unseen

Stack (top at right): D | F | J | I

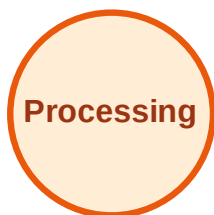
Visited: A, B, C, E



DFS Traversal

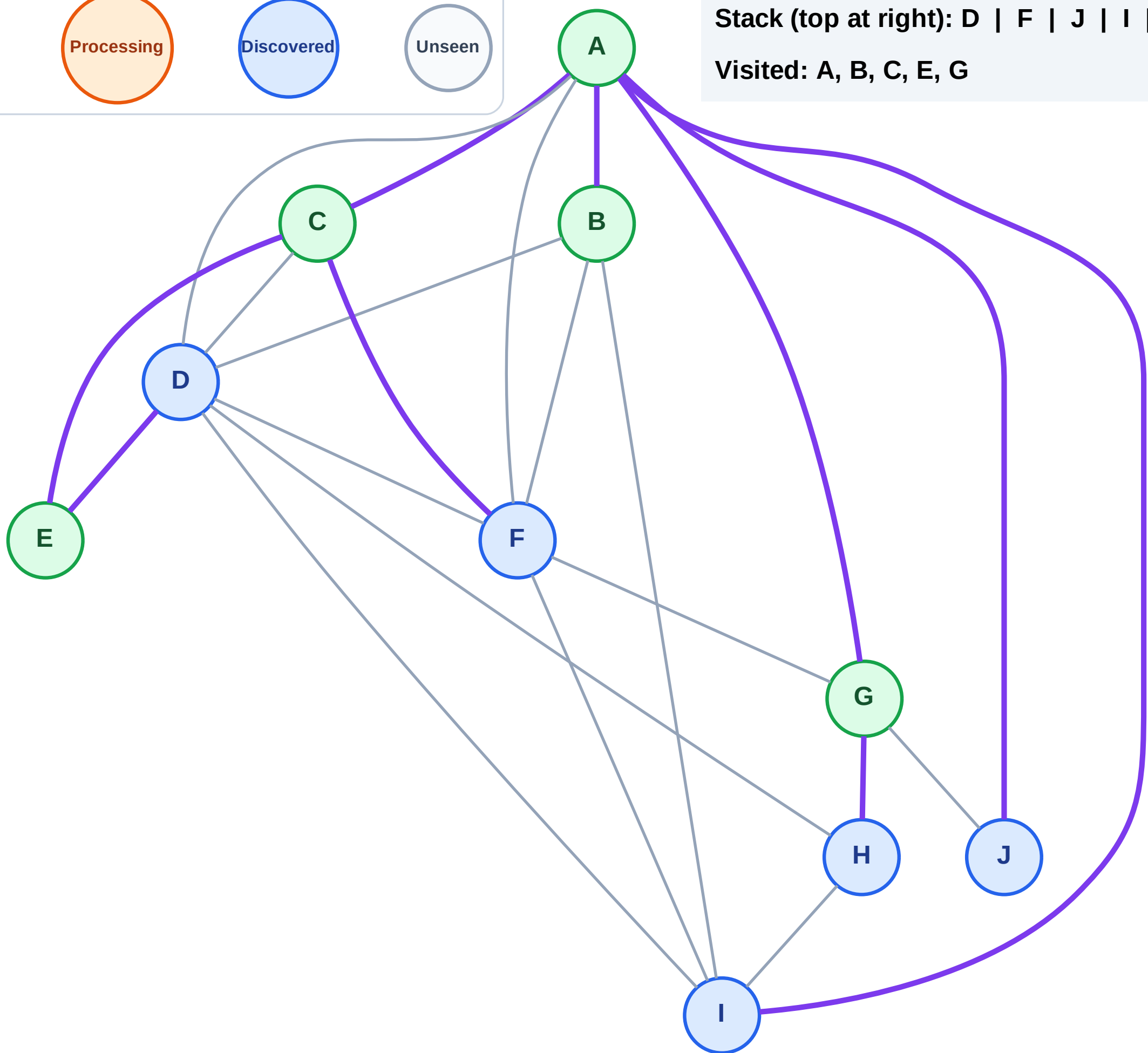
Step 11: Discover H from G

Legend



Stack (top at right): D | F | J | I | H

Visited: A, B, C, E, G



DFS Traversal

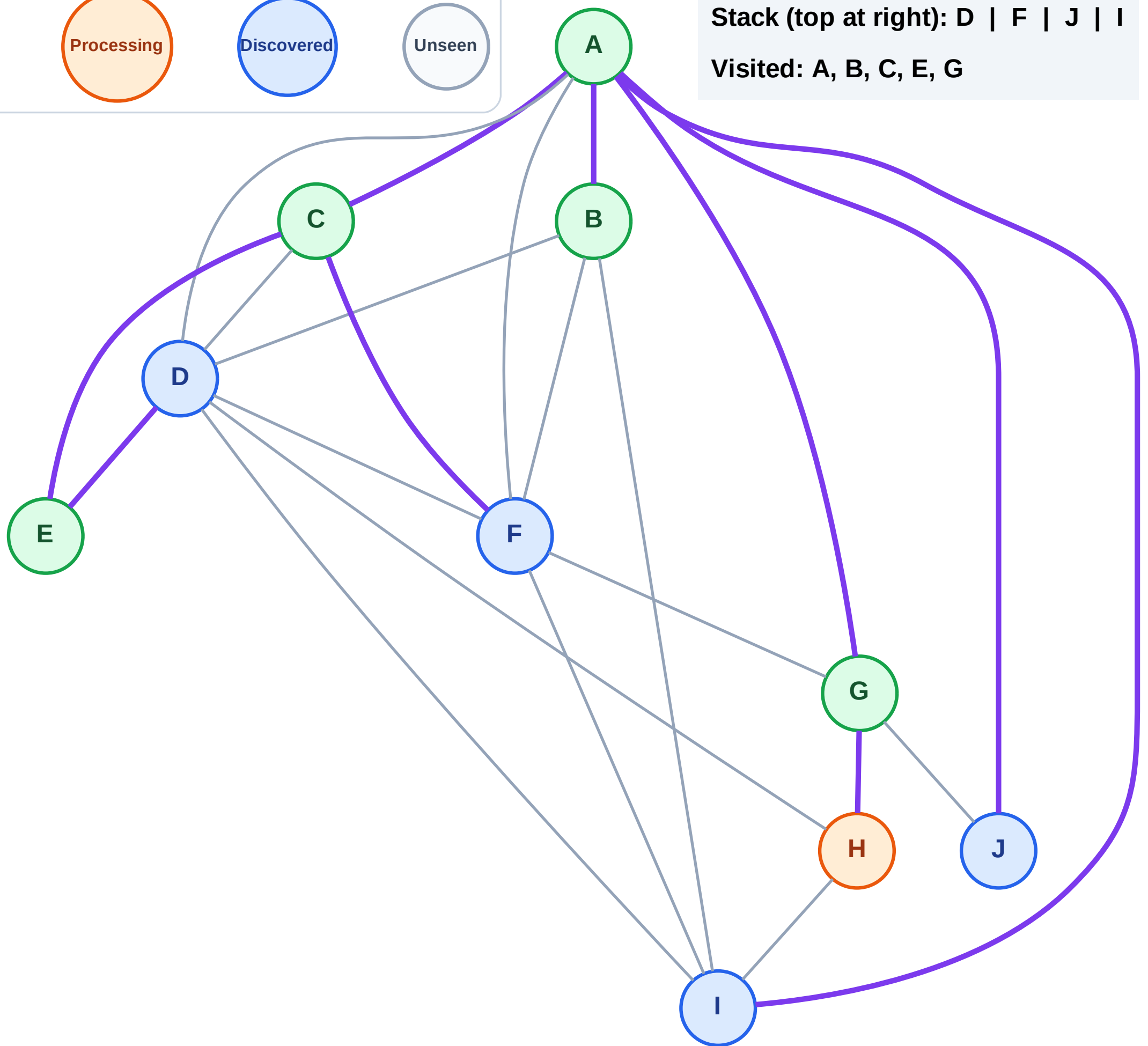
Step 12: Process node H

Legend



Stack (top at right): D | F | J | I

Visited: A, B, C, E, G



DFS Traversal

Step 13: No new neighbors from H

Legend

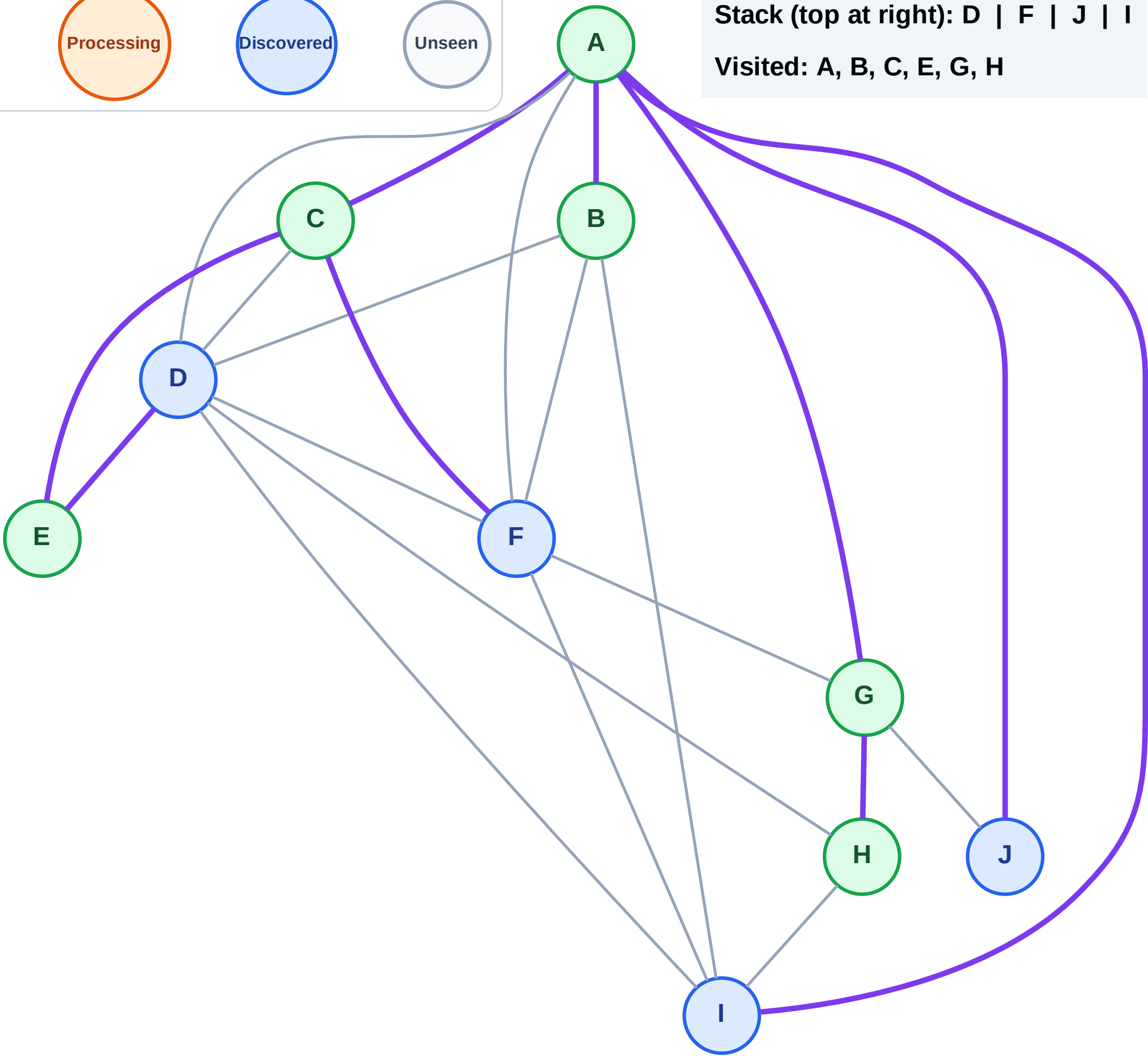
Complete

Processing

Discovered

Unseen

Stack (top at right): D | F | J | I
Visited: A, B, C, E, G, H



DFS Traversal

Step 14: Process node I

Legend

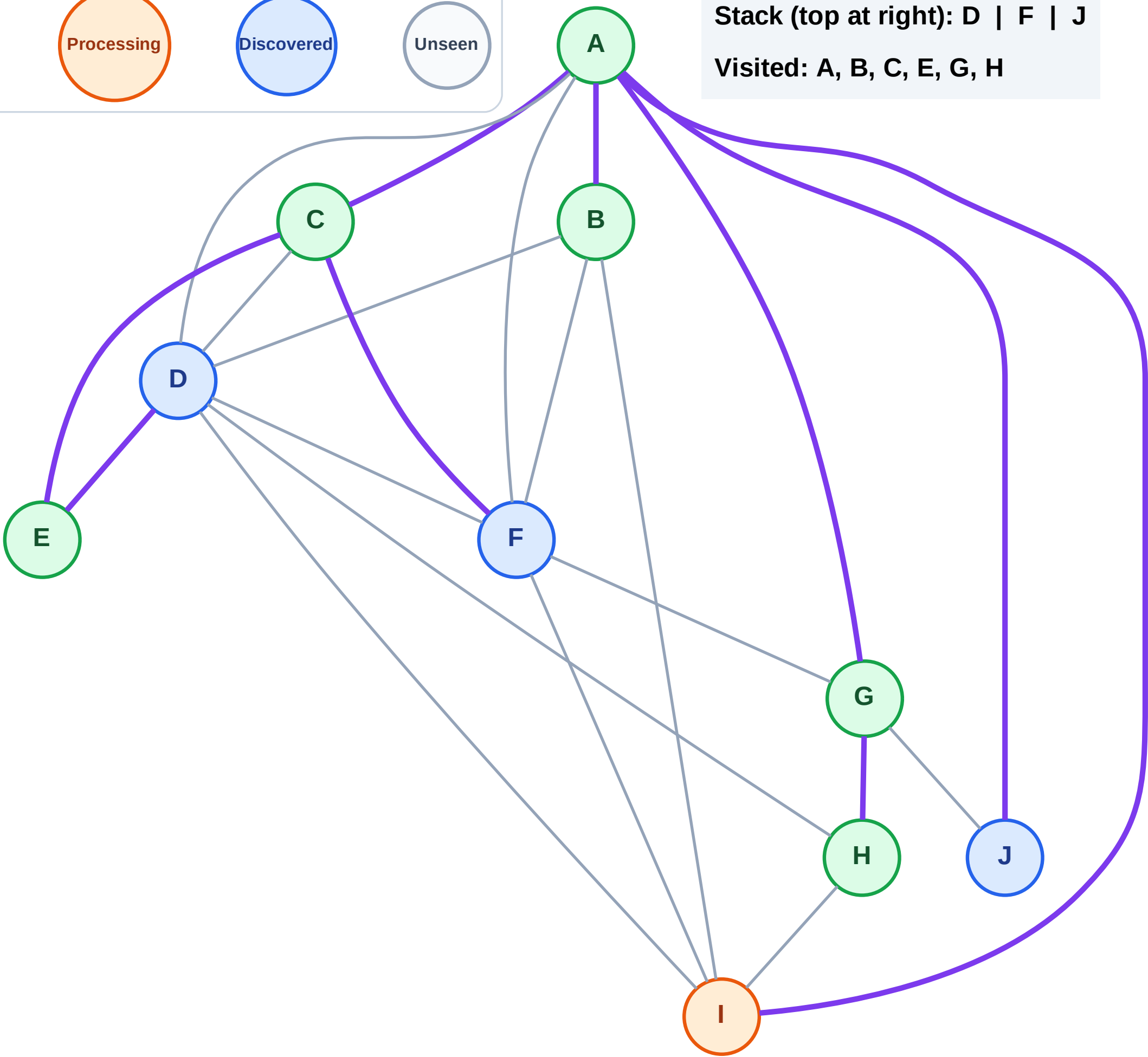
Complete

Processing

Discovered

Unseen

Stack (top at right): D | F | J
Visited: A, B, C, E, G, H



DFS Traversal

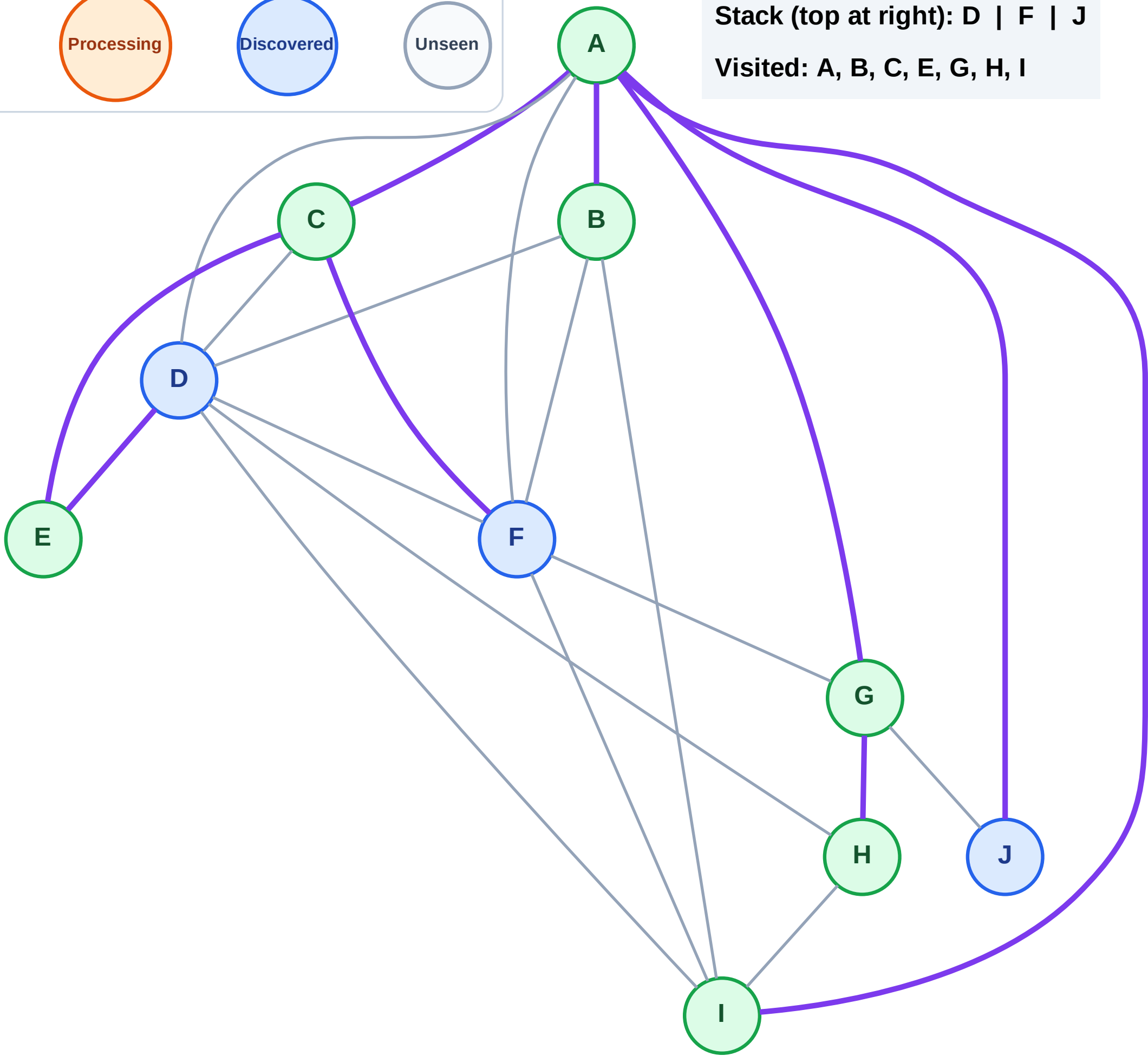
Step 15: No new neighbors from I

Legend



Stack (top at right): D | F | J

Visited: A, B, C, E, G, H, I



DFS Traversal

Step 16: Process node J

Legend

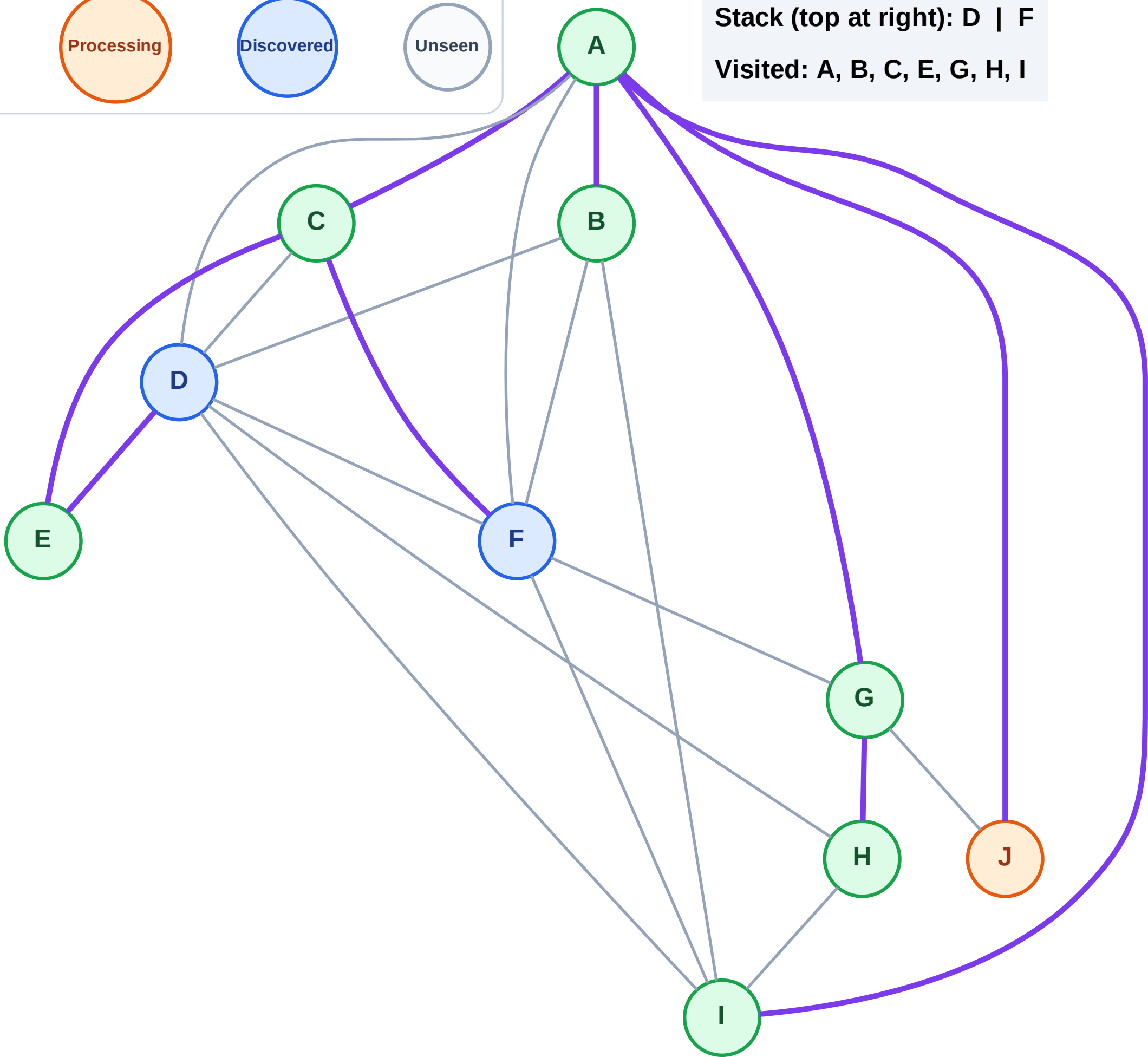
Complete

Processing

Discovered

Unseen

Stack (top at right): D | F
Visited: A, B, C, E, G, H, I



DFS Traversal

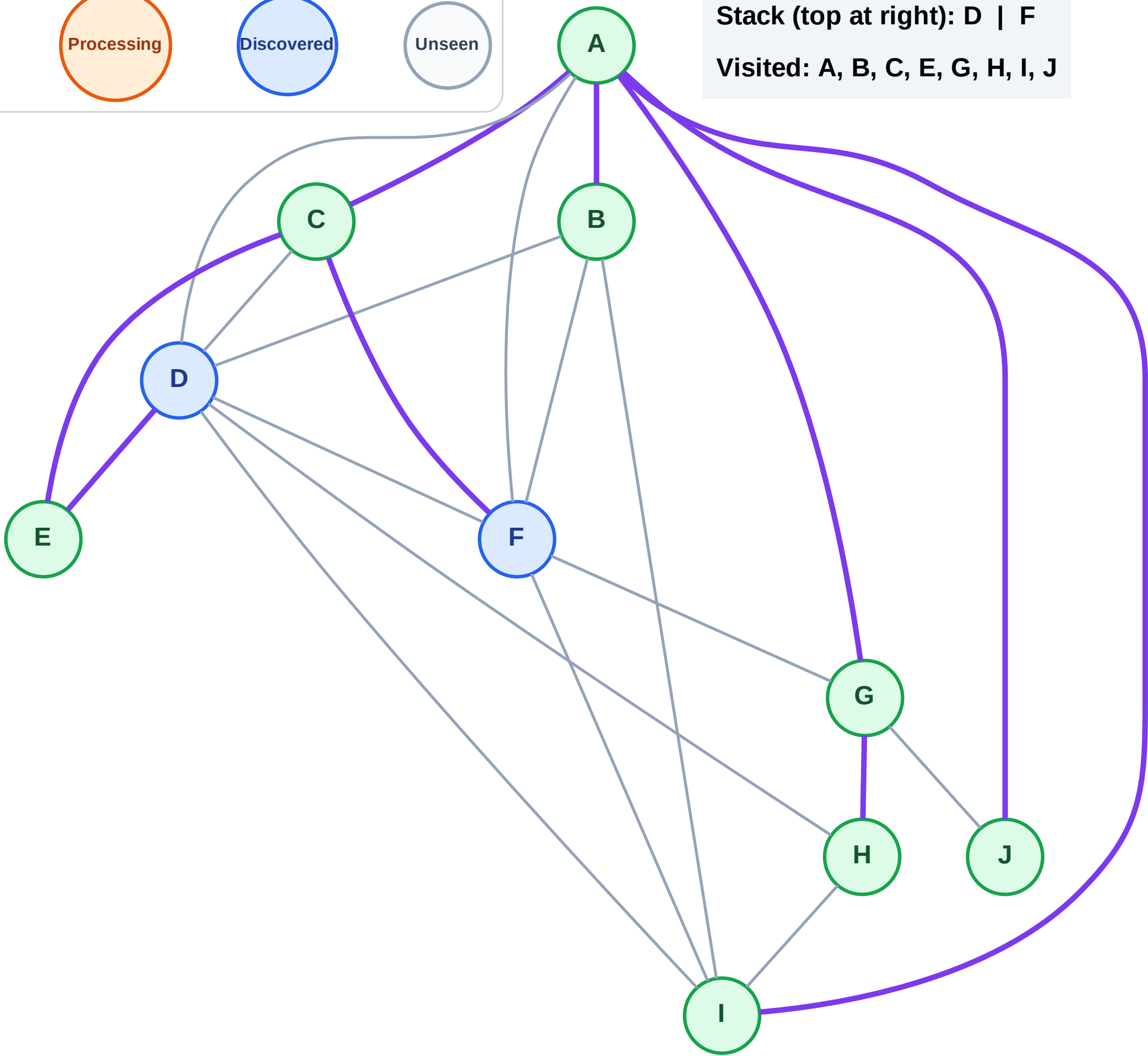
Step 17: No new neighbors from J

Legend



Stack (top at right): D | F

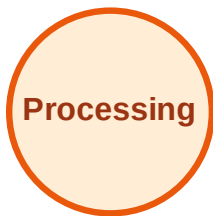
Visited: A, B, C, E, G, H, I, J



DFS Traversal

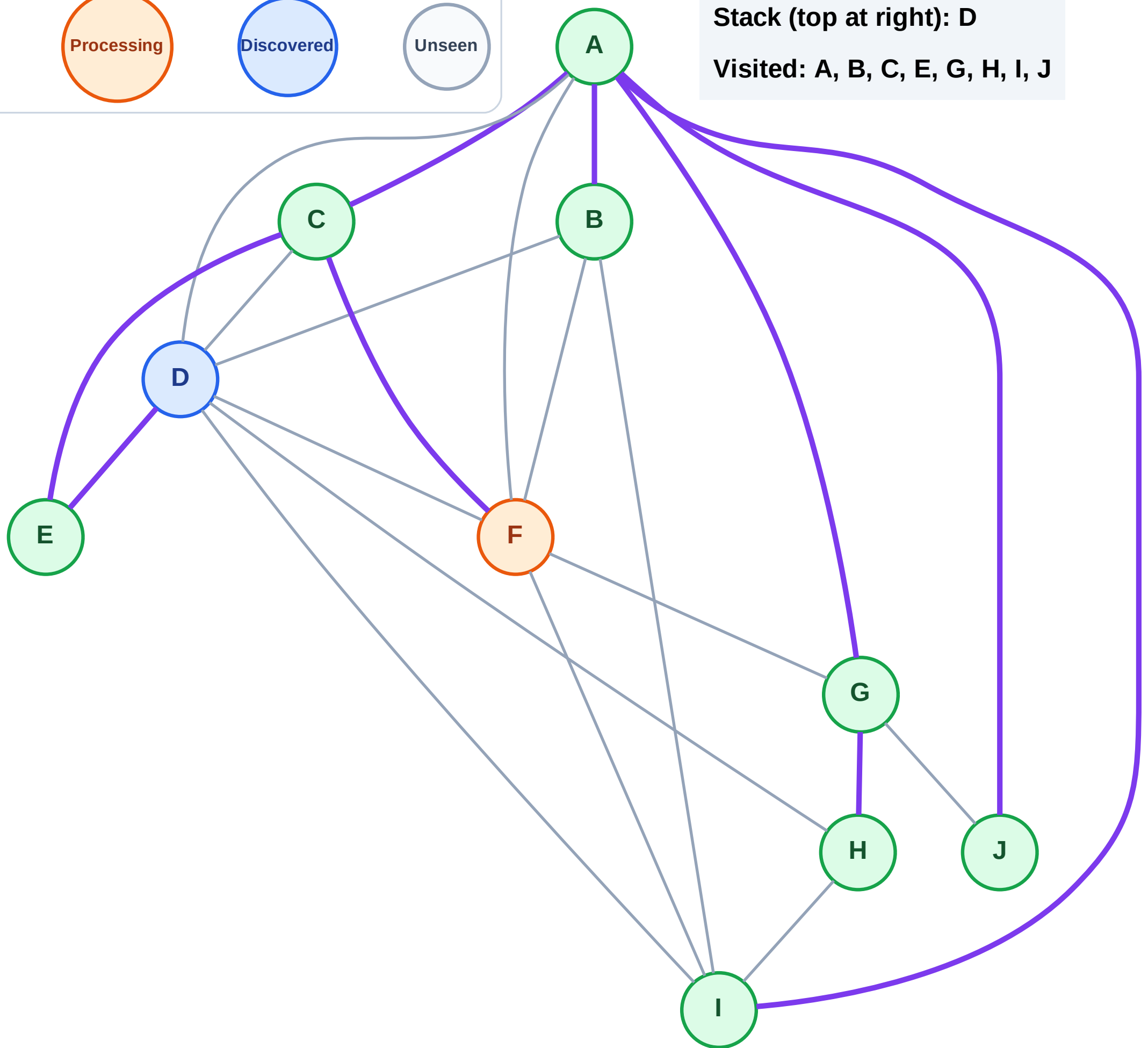
Step 18: Process node F

Legend



Stack (top at right): D

Visited: A, B, C, E, G, H, I, J



DFS Traversal

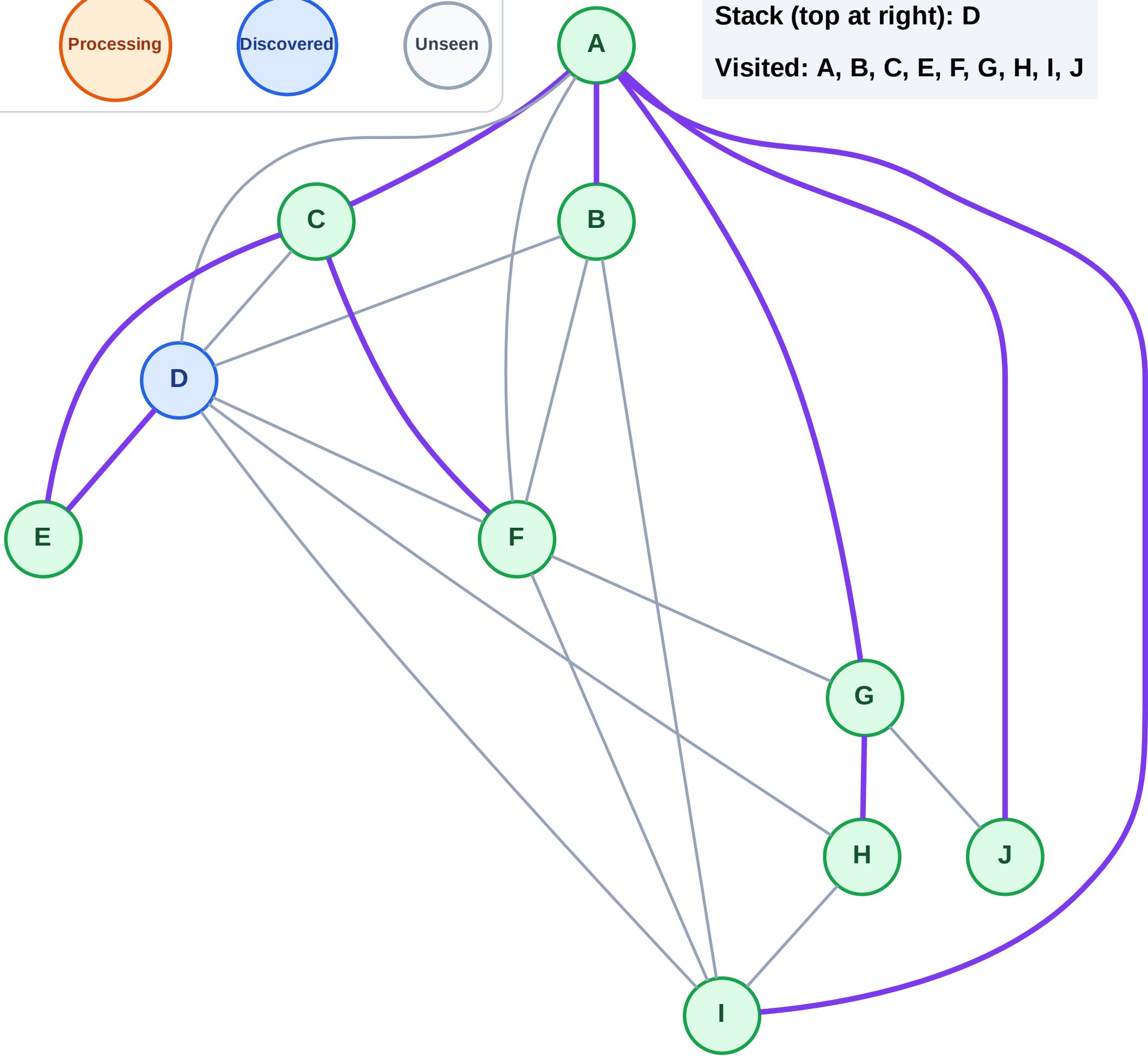
Step 19: No new neighbors from F

Legend



Stack (top at right): D

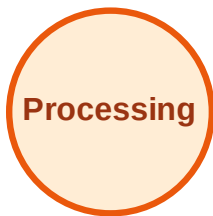
Visited: A, B, C, E, F, G, H, I, J



DFS Traversal

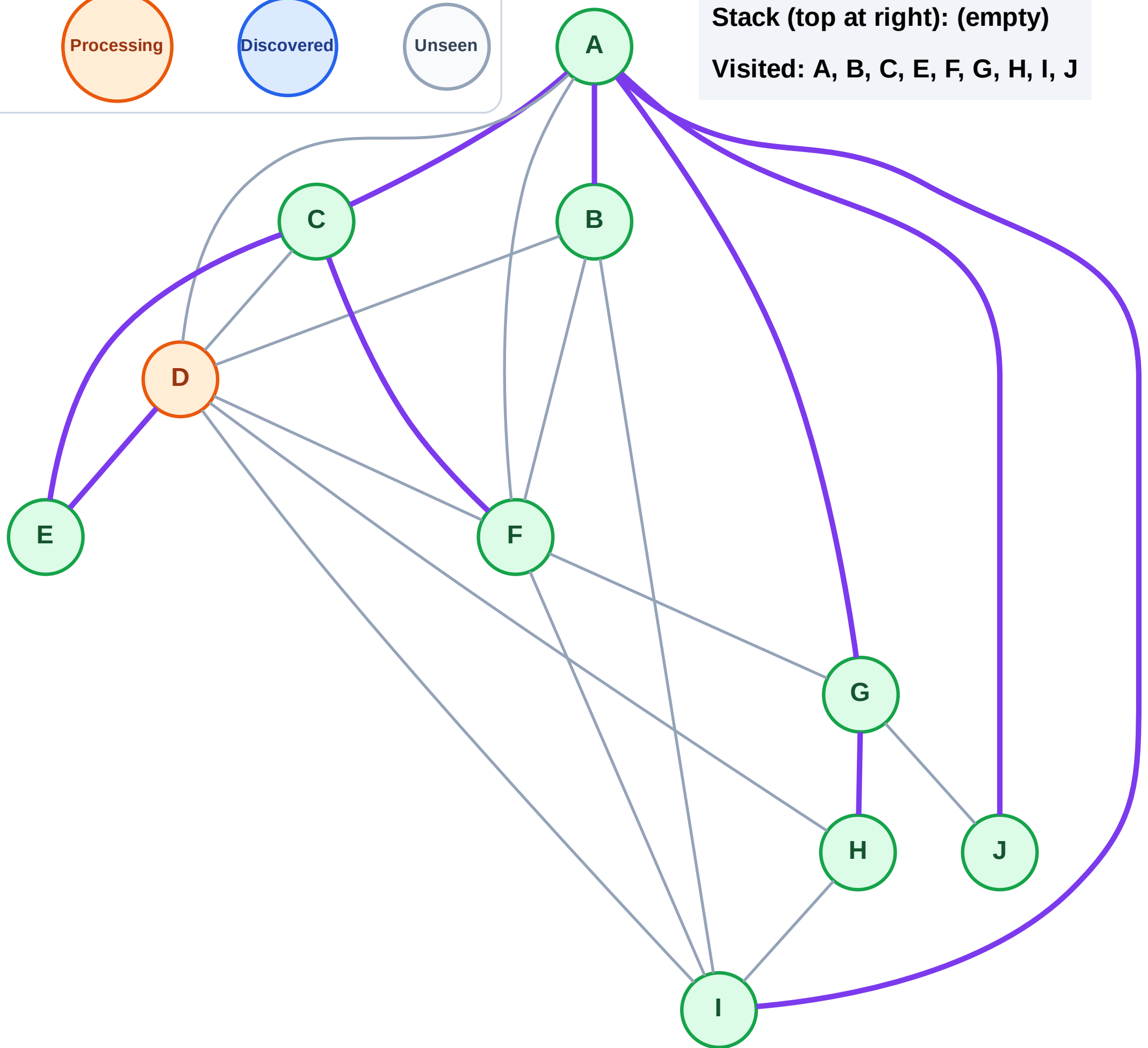
Step 20: Process node D

Legend



Stack (top at right): (empty)

Visited: A, B, C, E, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from D

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

